

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 225 factors, ordered by L–S CAGR.

HOW TO READ

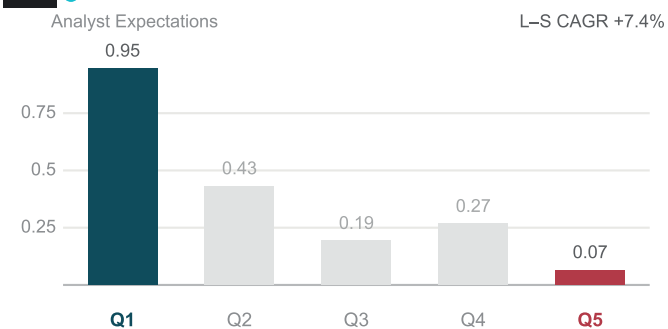
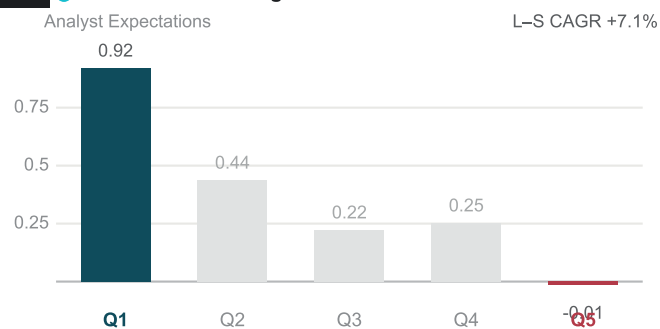
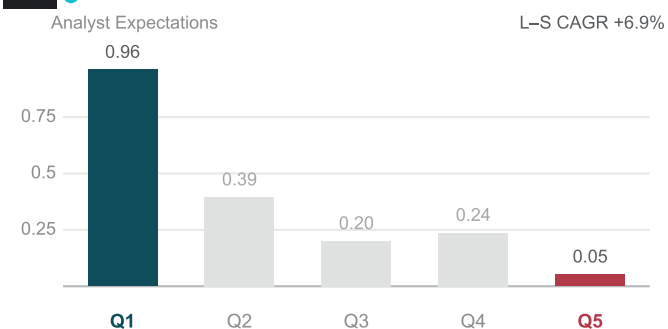
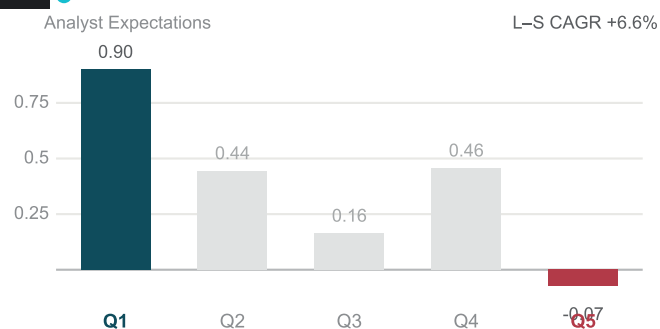
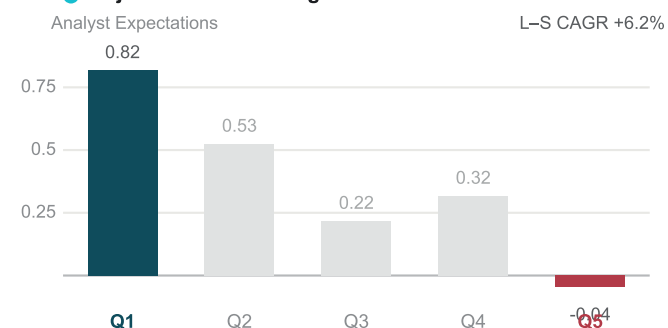
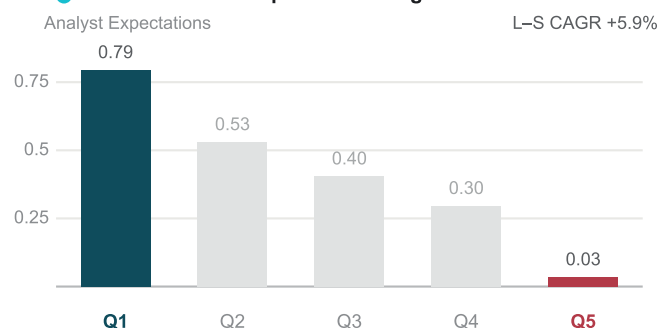
 Long  Short  Neutral

Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

FACTOR STYLES

 Valuation	35	 Price Momentum	50
 Earnings Quality	46	 Size	4
 Analyst Expectations	18	 Historical Growth	38
 Capital Efficiency	27	 Volatility	7

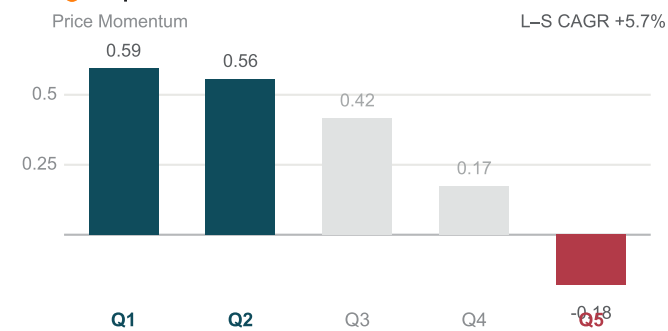
 In current model portfolio (28 factors)

001 3M Revision in FY2 EPS Estimate**002 Street Revision Magnitude****003 3M Revision in FY1 EPS Estimate****004 Number of EPS FY1 Revisions****005 Adjusted Revision Magnitude****006 Standardized Unexpected Earnings**

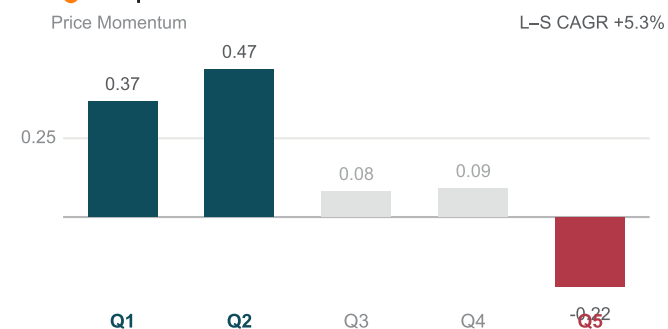
007 ● 6M Chg in Target Price



008 ● Slope of 52W Price Trend



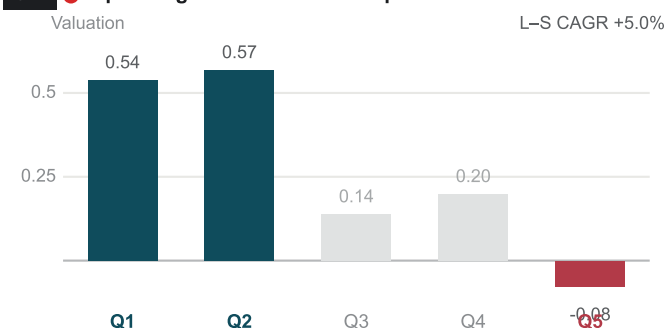
009 ● Sharpe Ratio



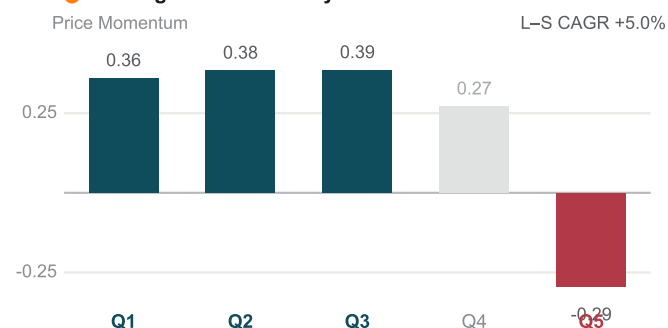
010 ● Cash Flow Return on Invested Capital



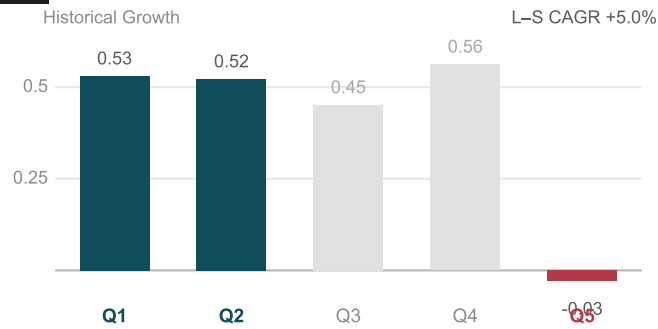
011 ● Operating Cash Flow to Enterprise Value



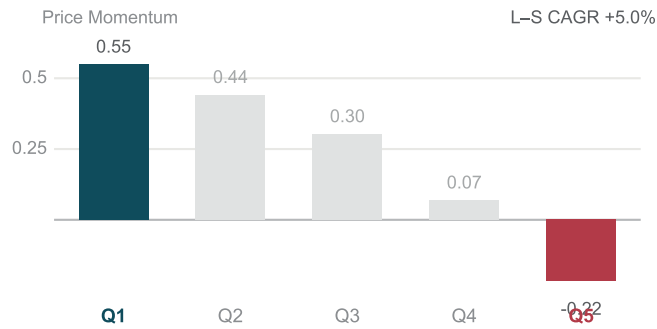
012 ● Closing Price to 260 Day Low



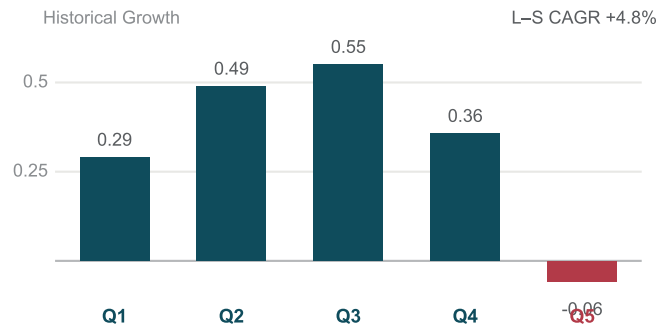
013 Dividend per Share Growth



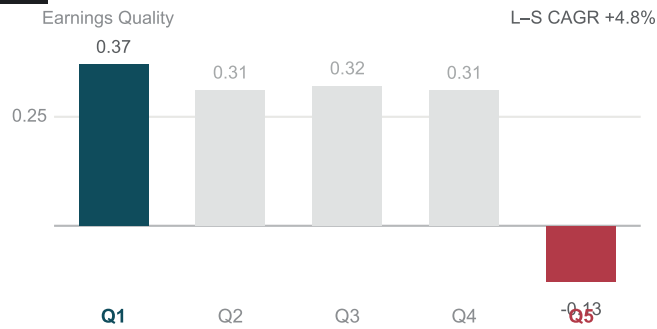
014 24M Residual Return Variance



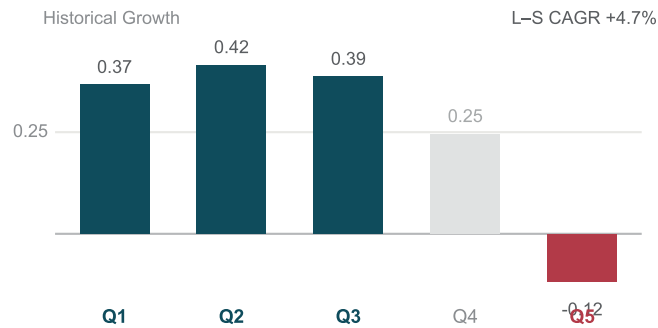
015 3Y Chg in Earnings per Share



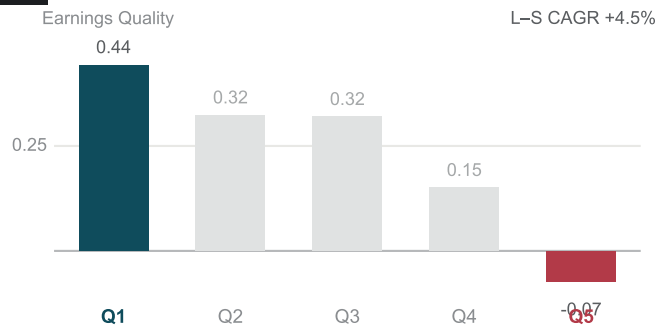
016 ROE 20 Qtr Standard Deviation



017 1Y Chg in Sales Turnover



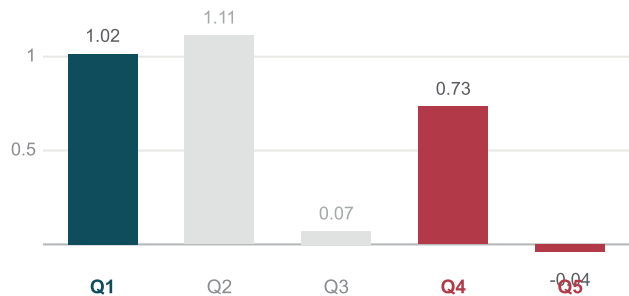
018 ROA 20 Qtr Standard Deviation



019 ● **Adjusted Number of EPS FY1 Revisions**

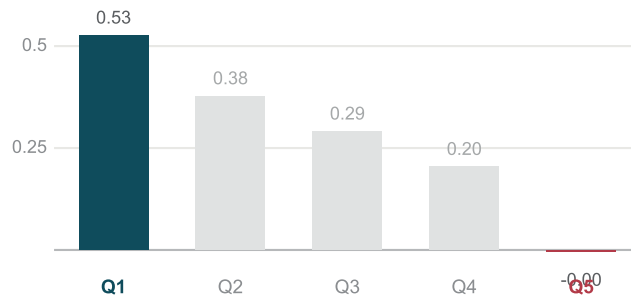
Analyst Expectations

L-S CAGR +4.5%

020 ● **Operating Cash Flow to Equity**

Capital Efficiency

L-S CAGR +4.4%

021 ● **Free Cash Flow to Sales**

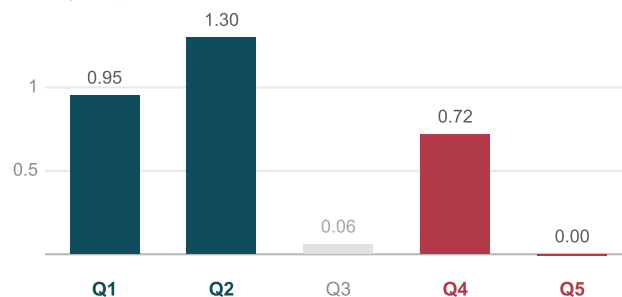
Earnings Quality

L-S CAGR +4.3%

022 ● **Analyst Earnings Estimate Diffusion**

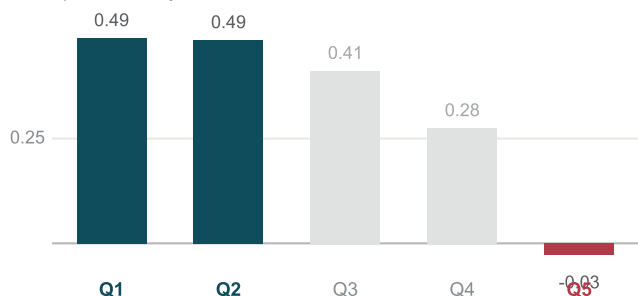
Analyst Expectations

L-S CAGR +4.3%

023 ● **Operating Cash Flow Ratio**

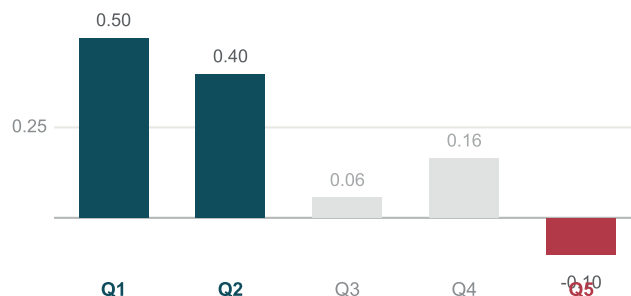
Capital Efficiency

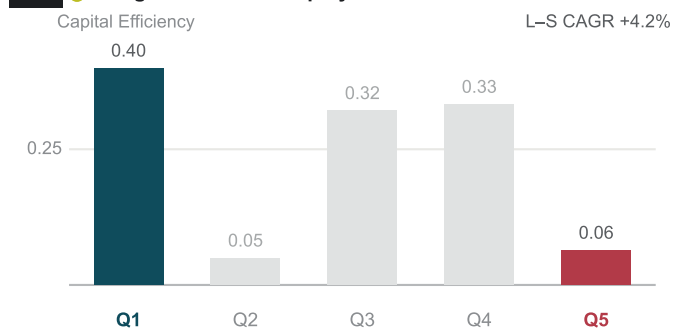
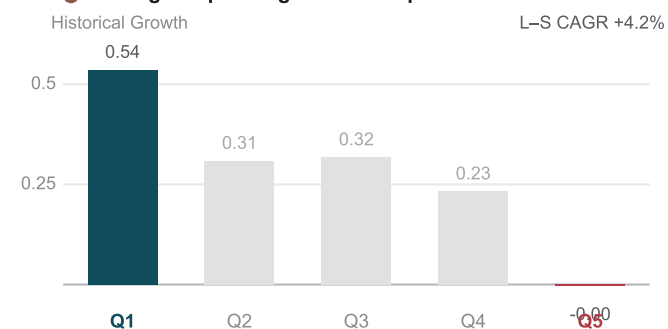
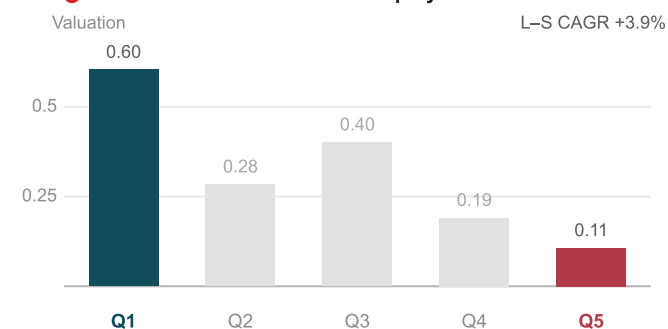
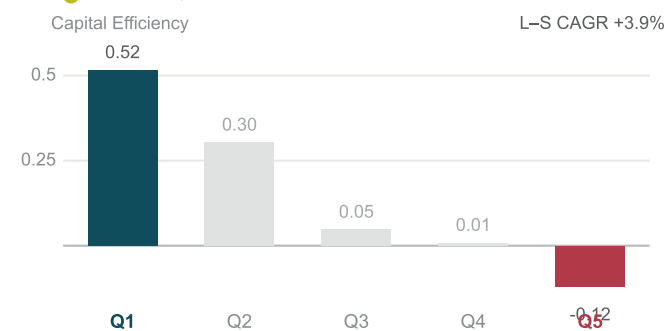
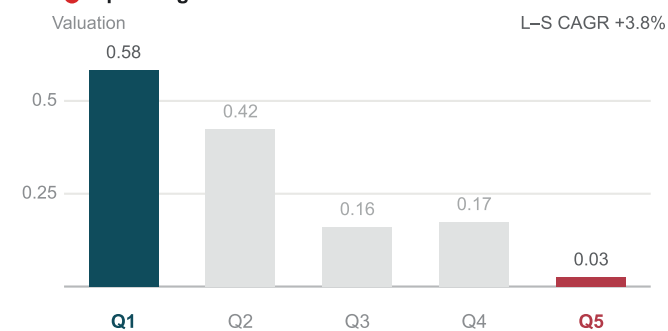
L-S CAGR +4.3%

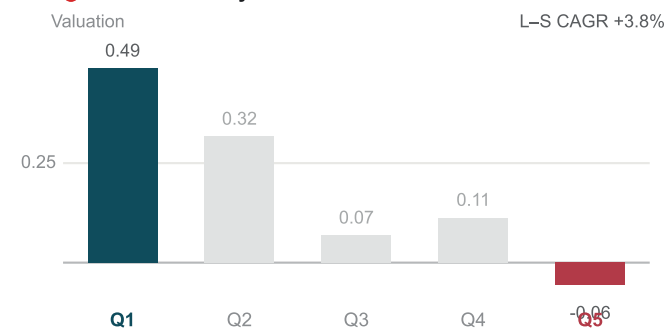
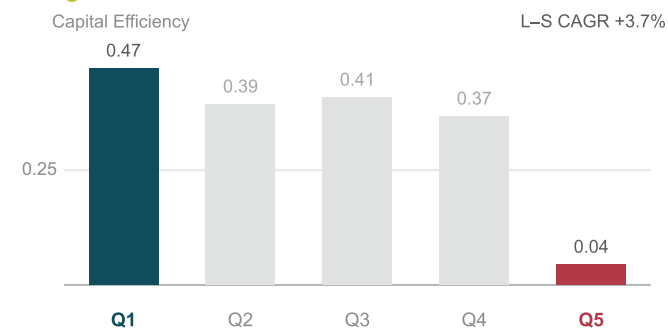
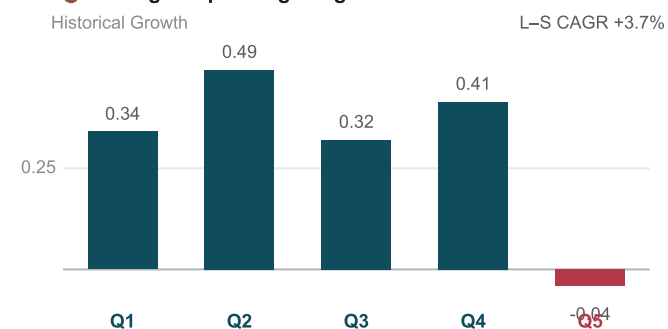
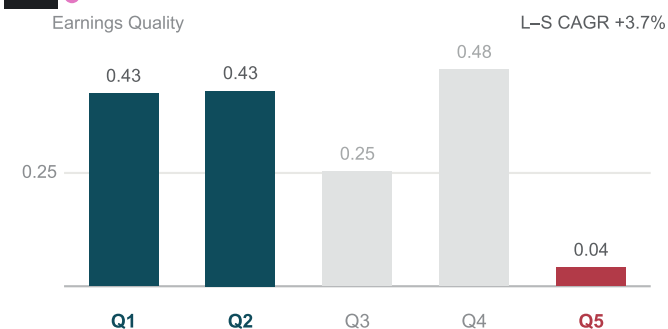
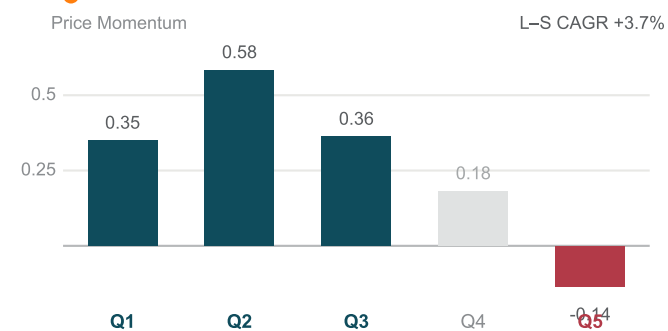
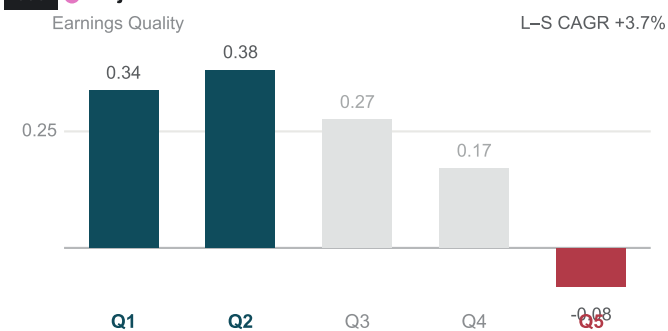
024 ● **Sales to Price Ratio**

Valuation

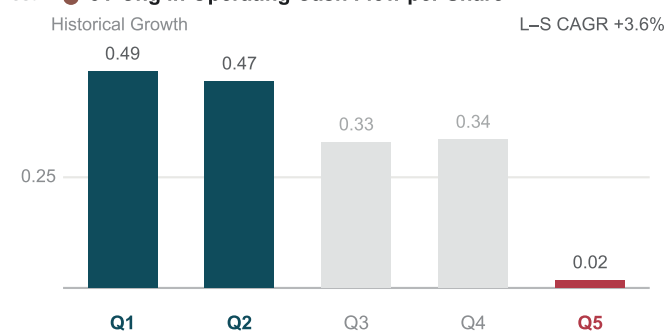
L-S CAGR +4.3%



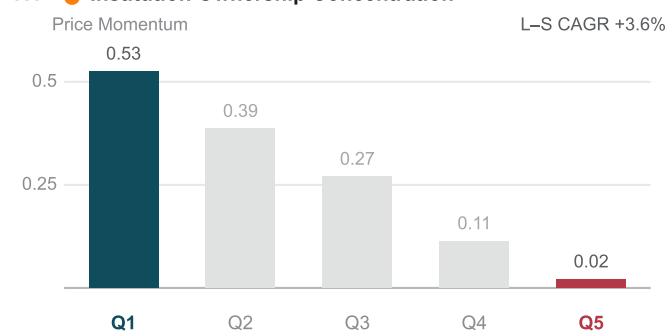
025 ● **Net Profit Margin**026 ● **Long Term Debt to Equity Ratio**027 ● **1Y Chg in Operating Cash Flow per Share**028 ● **Free Cash Flow to Common Equity**029 ● **Tobin's Q Ratio**030 ● **Operating Cash Flow to Price**

031 ● **Current Liability to Price**032 ● **Gross Profit to Assets**033 ● **3Y Chg in Operating Margin**034 ● **Asset Turnover**035 ● **180D Price TStat**036 ● **Adjusted Accruals**

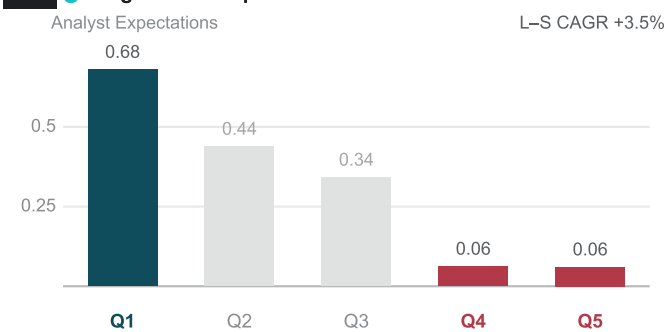
037 ● 3Y Chg in Operating Cash Flow per Share



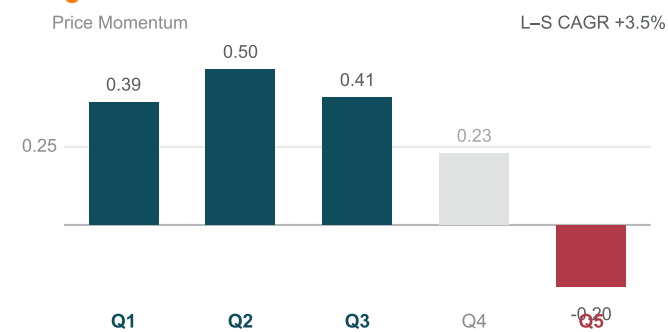
038 ● Institution Ownership Concentration



039 ● Target Price Gap to 6M EMA



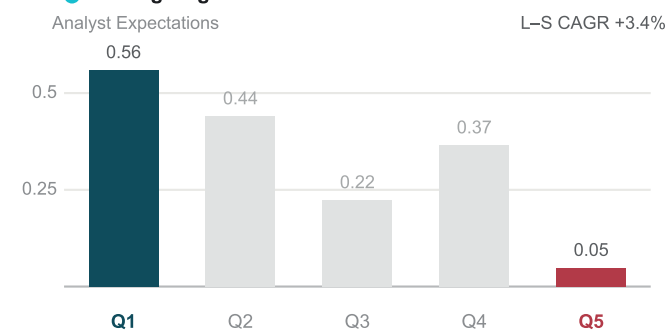
040 ● Share Turnover



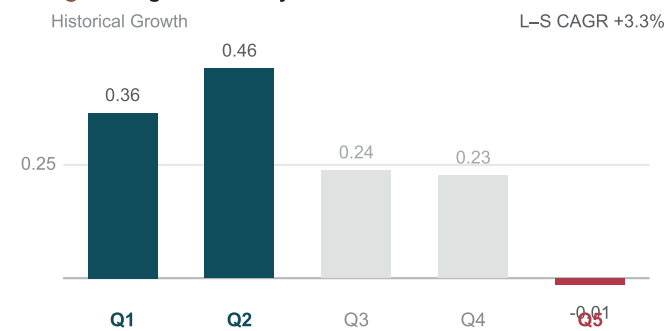
041 ● 1Y Chg in Sales to Price



042 ● 6M Avg Chg 1M Recommendation



043 ● 1Y Chg in Asset Adjusted EPS



044 ● 3Y Chg in Free Cash Flow per Share



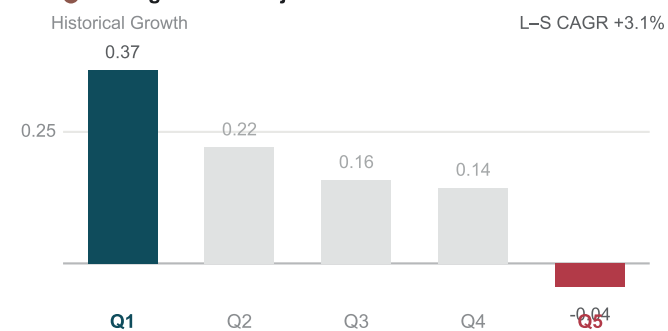
045 ● Capital Acquisition Ratio



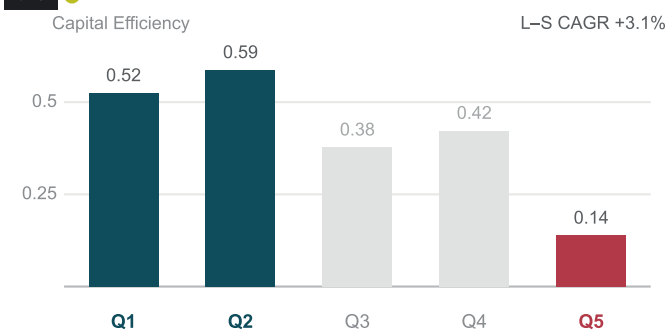
046 ● EBIT Margin

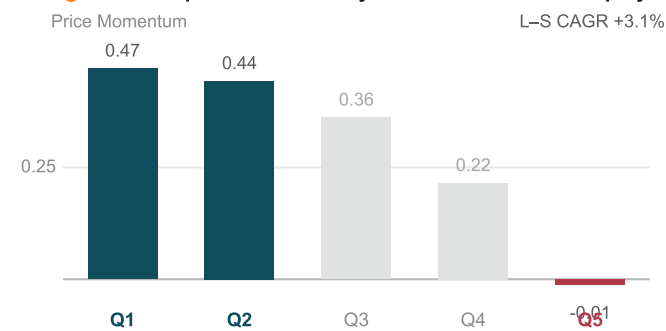
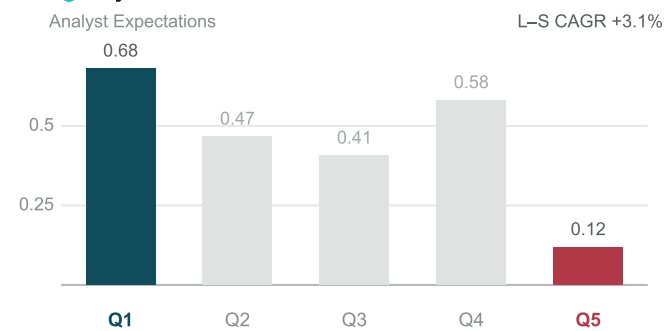
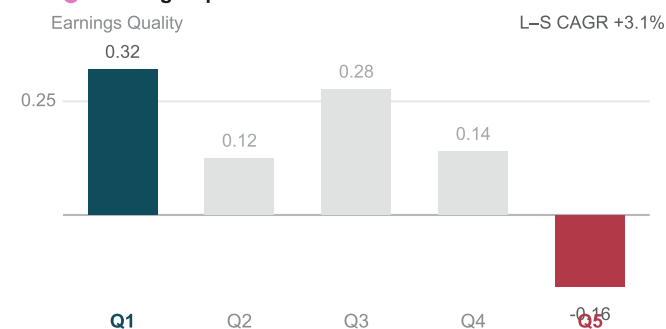
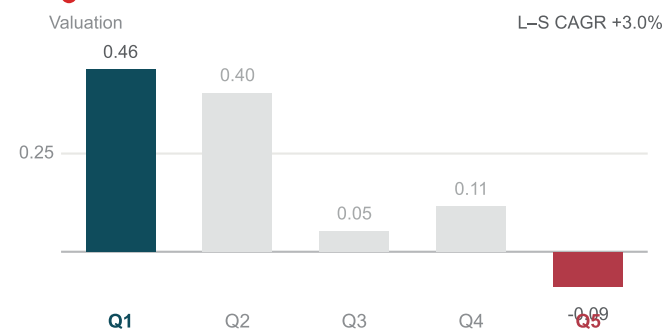
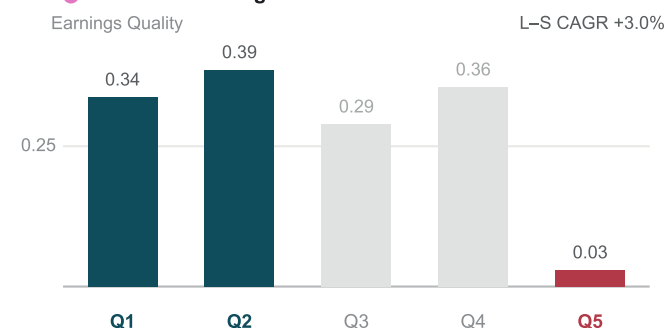
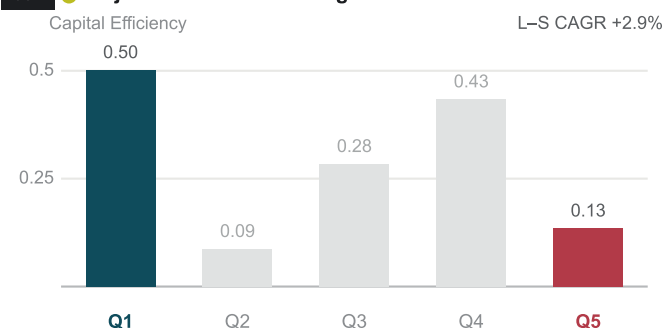


047 ● 1Y Chg in Asset Adjusted Free Cash Flow

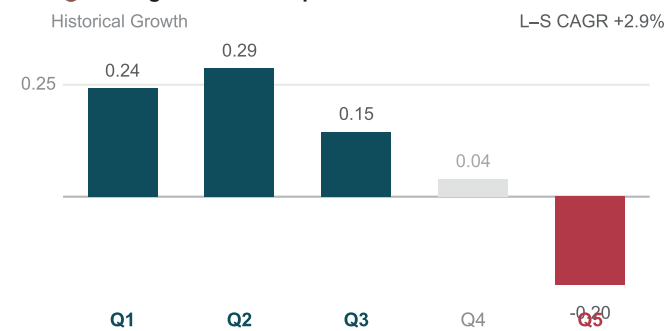


048 ● Receivables Turnover Ratio



049 ● **Ownership Breadth Stability of Institution Ownership by Hol...**050 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**051 ● **Working Capital Accruals**052 ● **Assets to Price Ratio**053 ● **Gross Profit Margin**054 ● **Adjusted Interest Coverage Ratio**

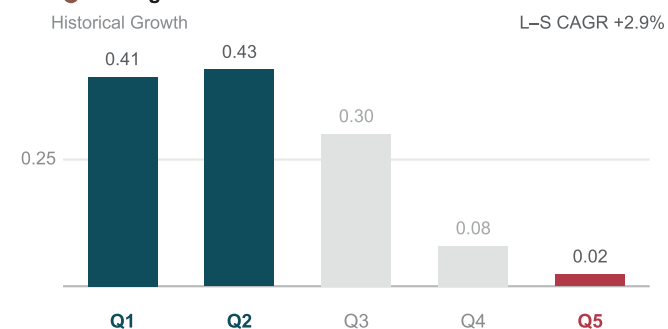
055 ● 1Y Chg in Cash Flow per Share



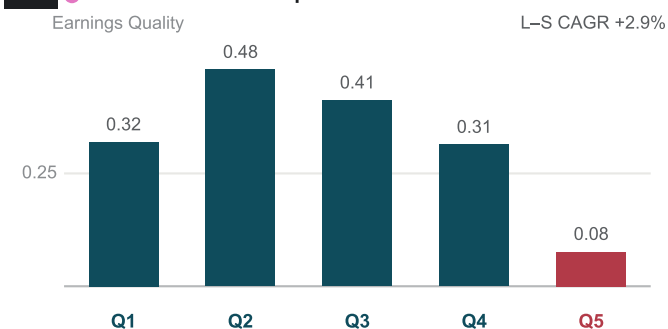
056 ● 1Y Chg in Free Cash Flow per Share



057 ● 1Y Chg in Cash Flow to Price



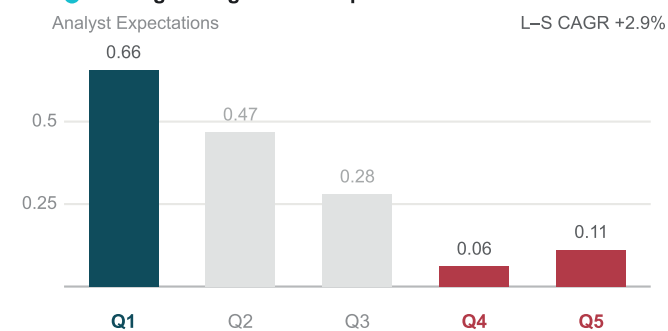
058 ● Sales to Invested Capital

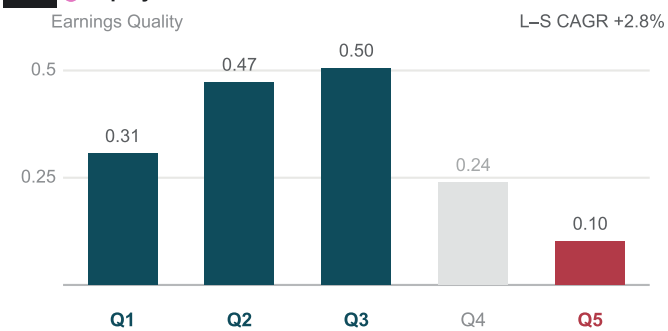
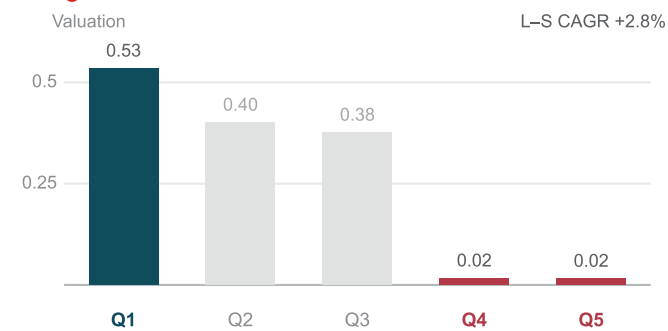
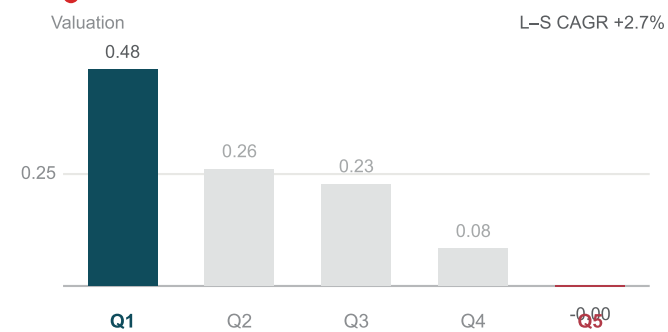
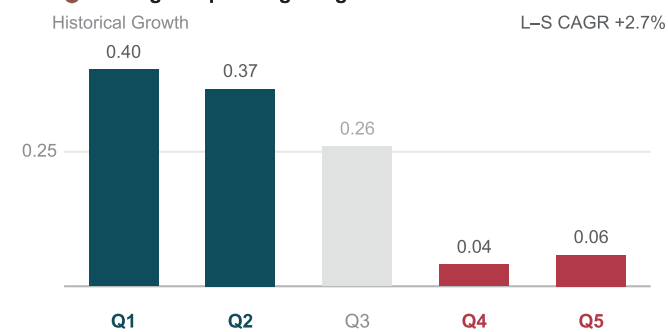
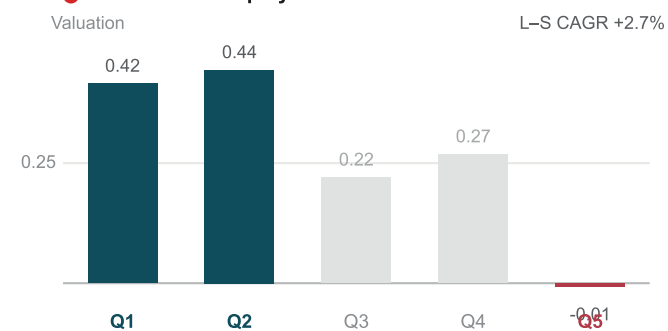
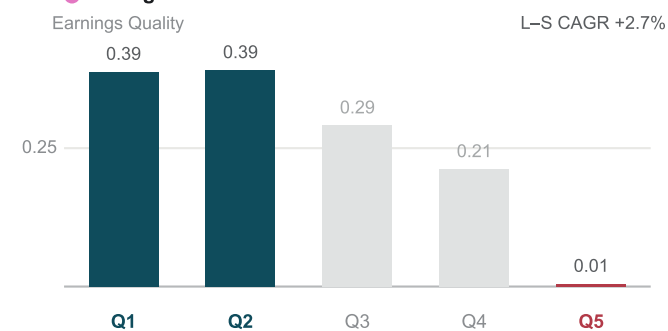


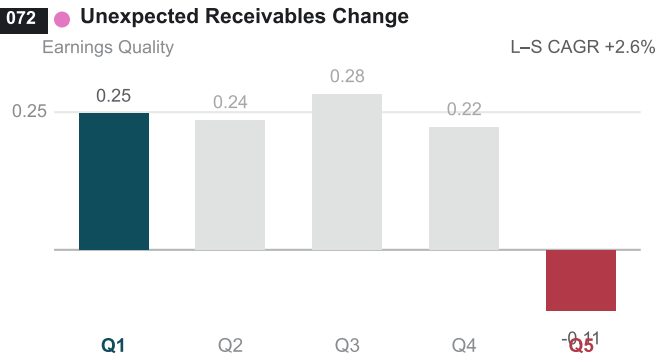
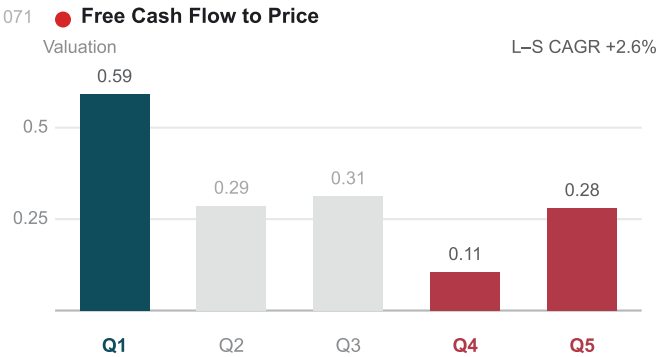
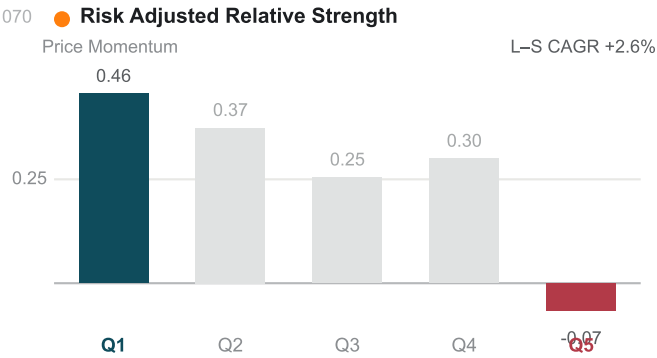
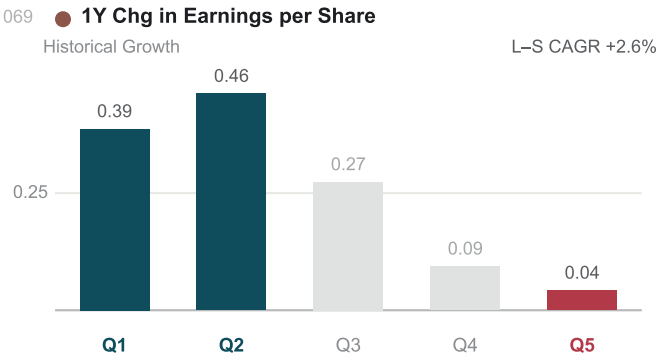
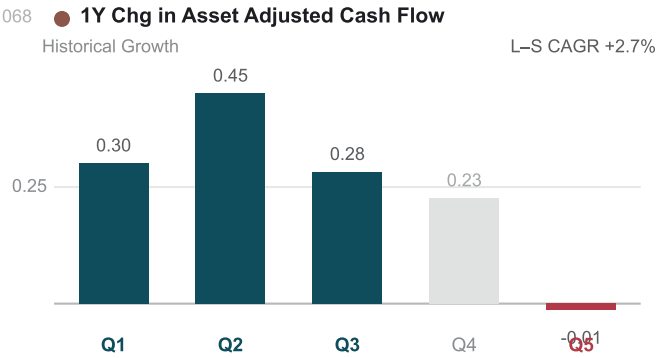
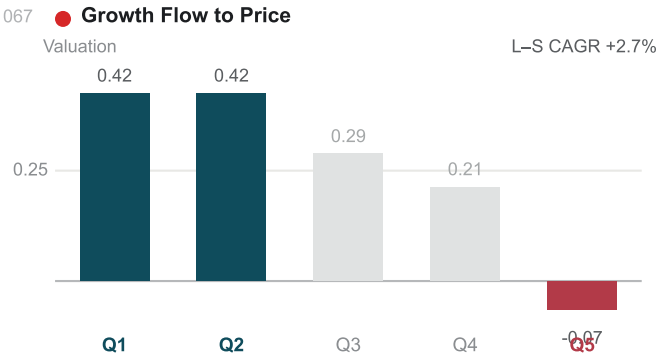
059 ● 1Y Chg in Earnings to Price

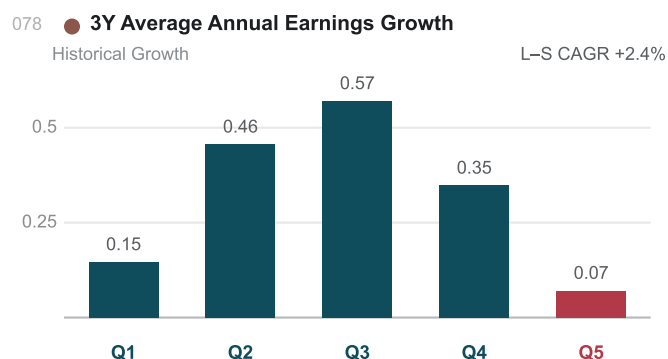
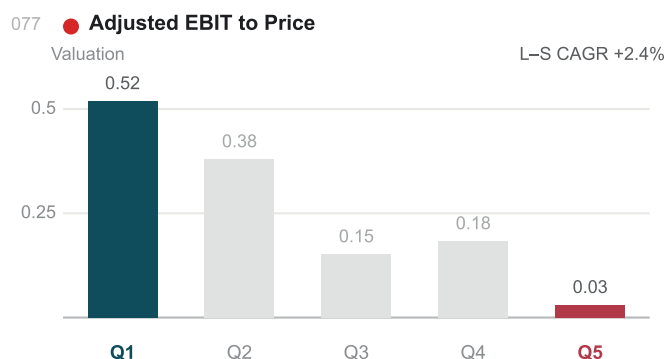
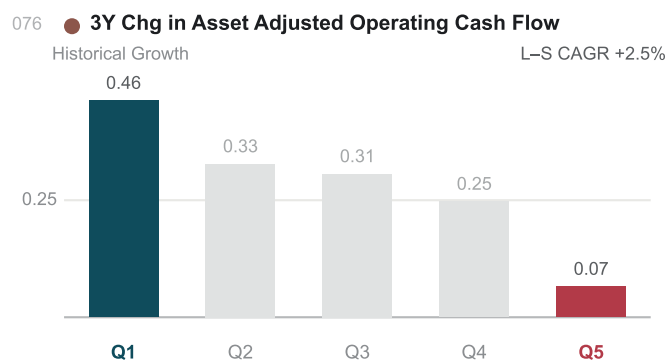
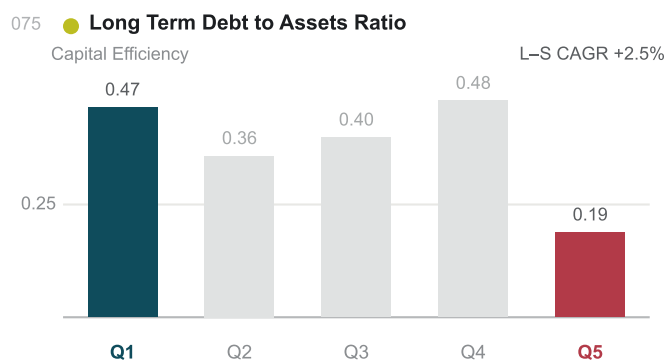
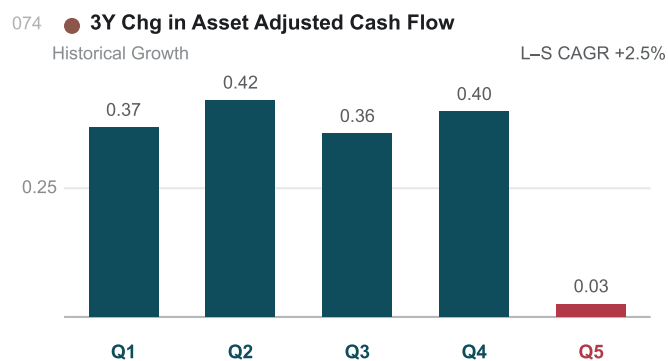
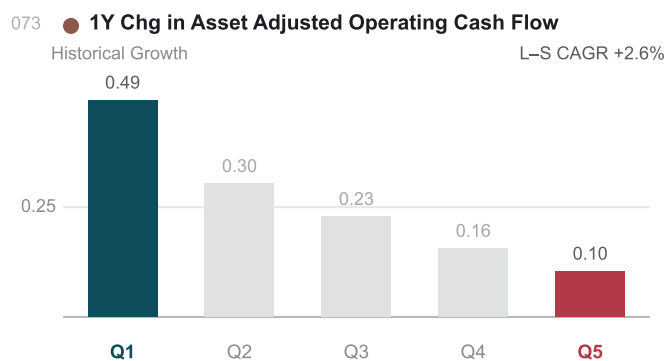


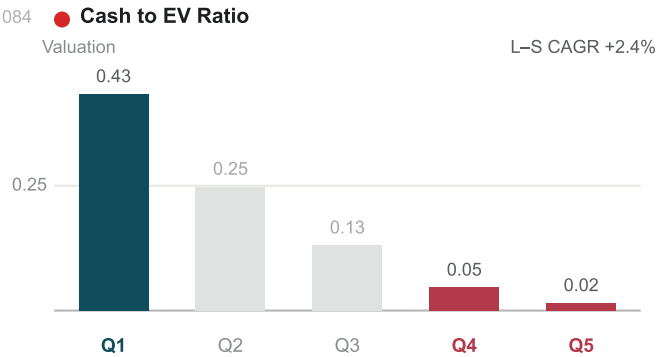
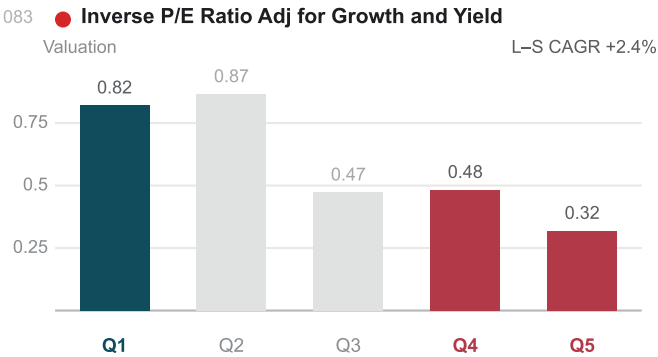
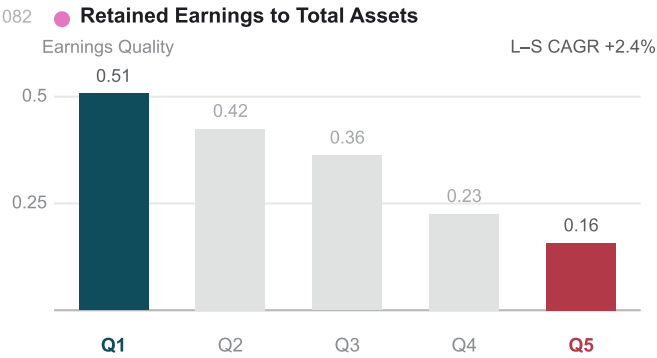
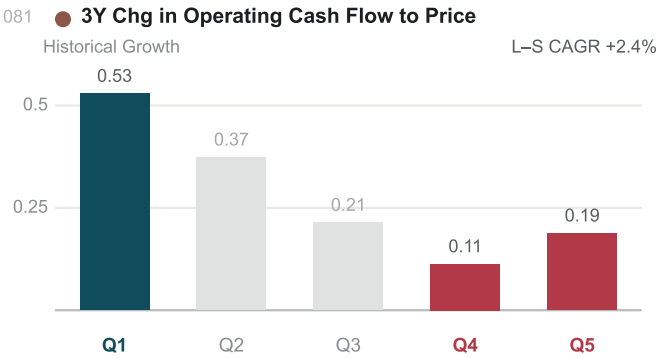
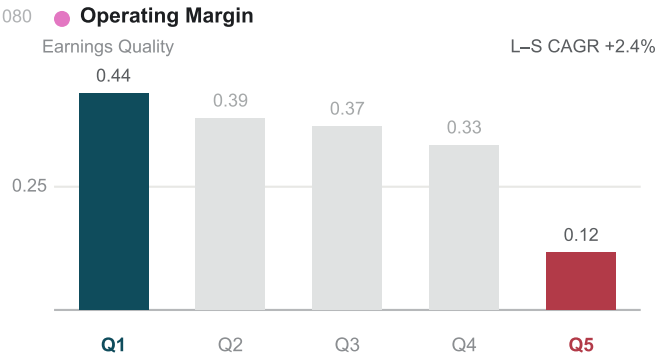
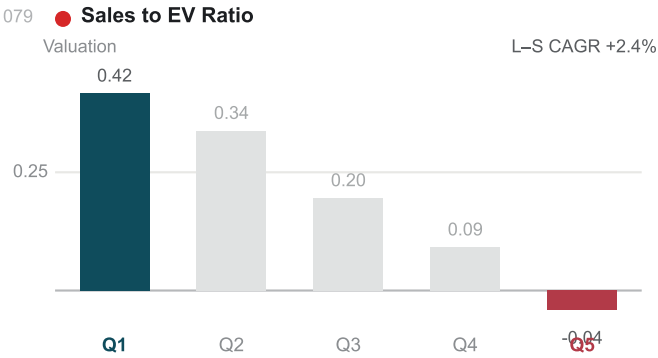
060 ● 6M Chg in Target Price Gap

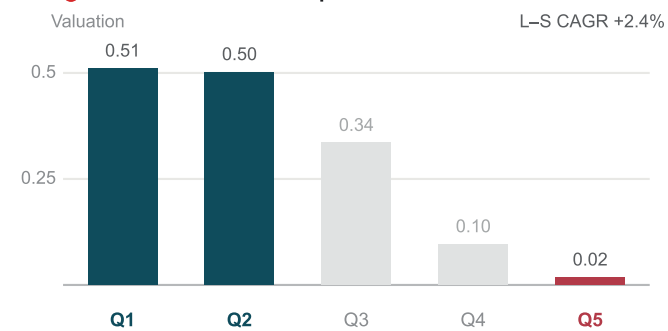
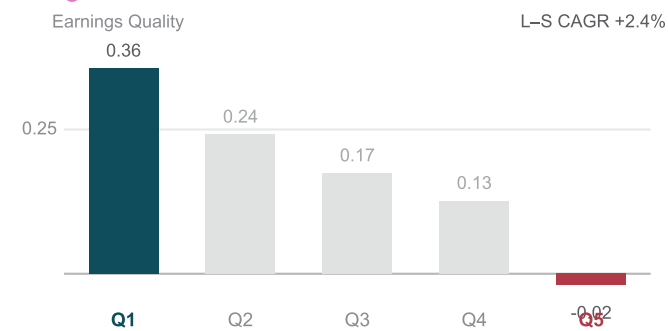
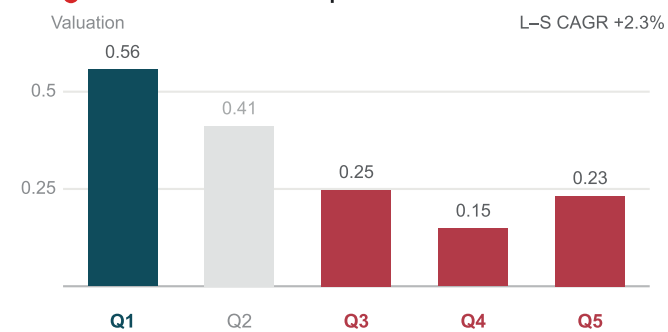
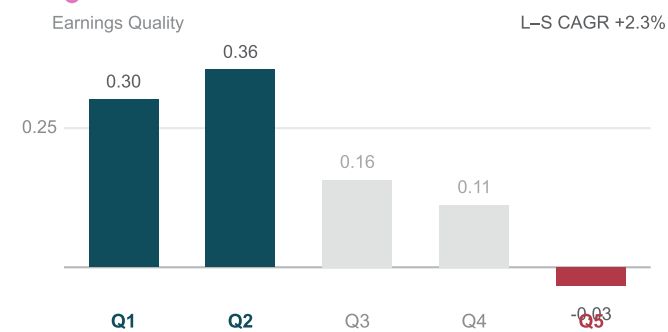
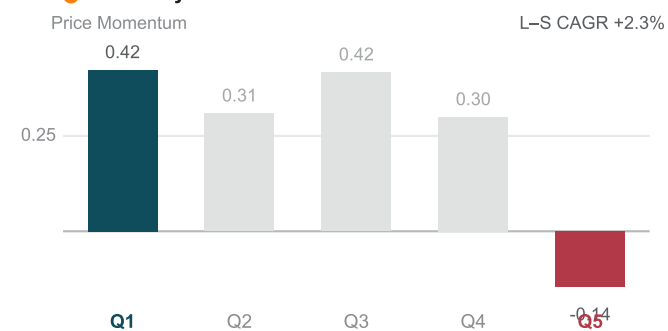
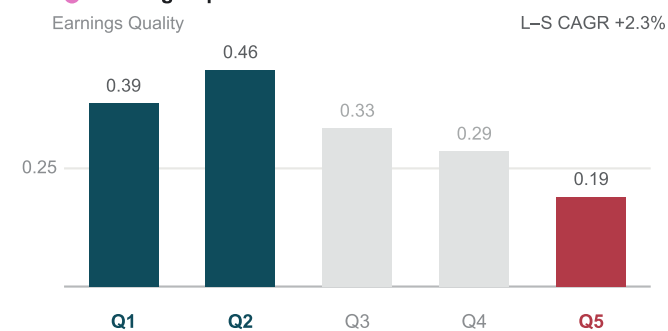


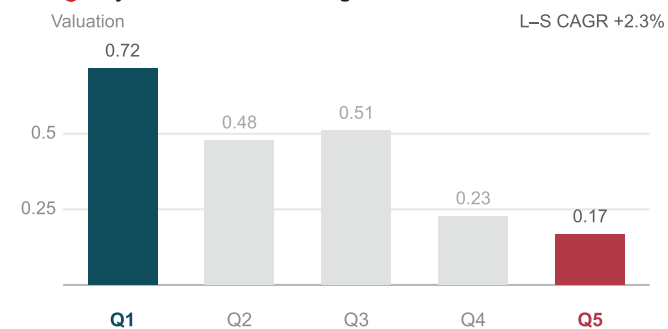
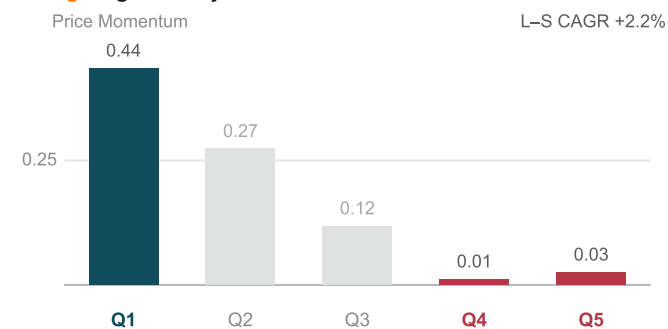
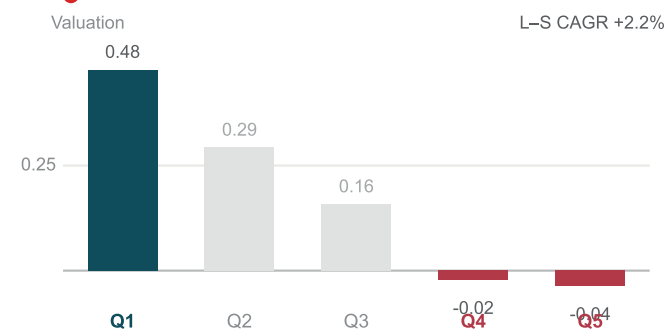
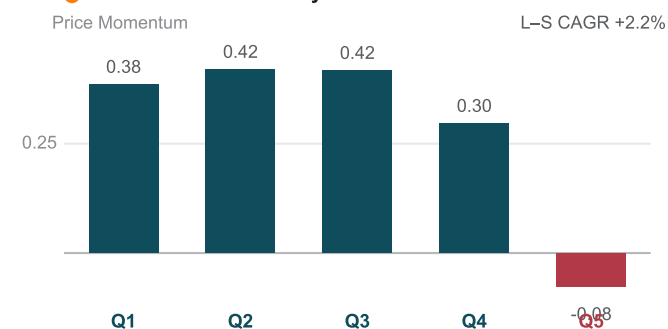
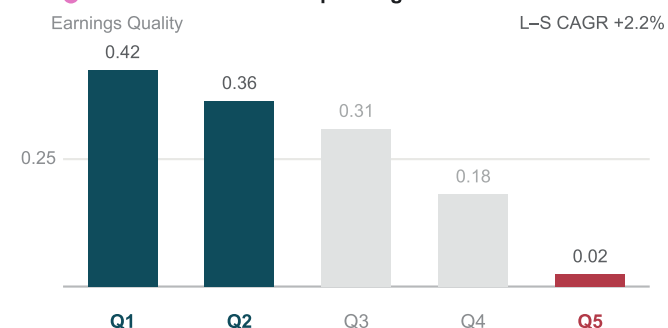
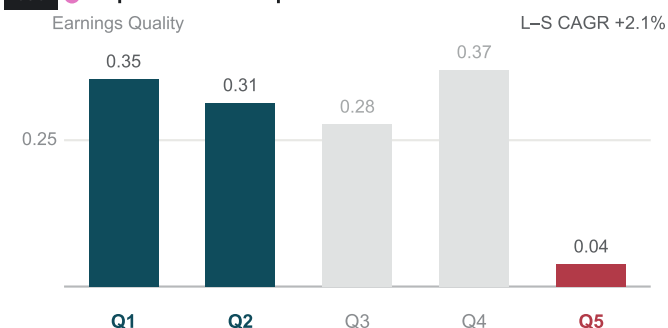
061 Equity Turnover

062 Net Cash Flow to Price

063 Cash to Price

064 1Y Chg in Operating Margin

065 Cash Flow to Equity

066 Change EPS to Sales


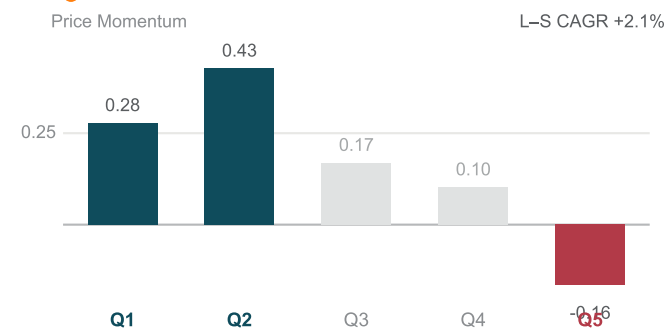
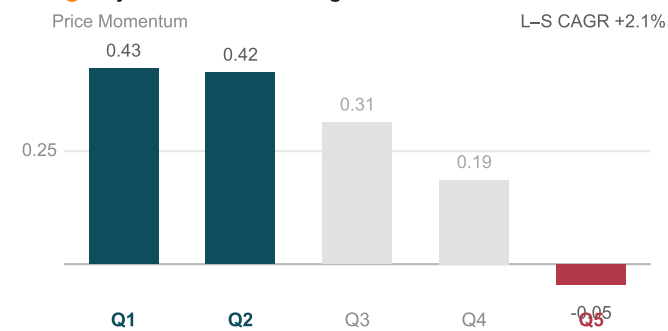
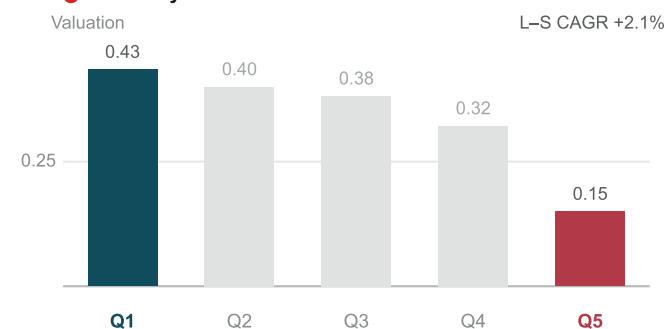
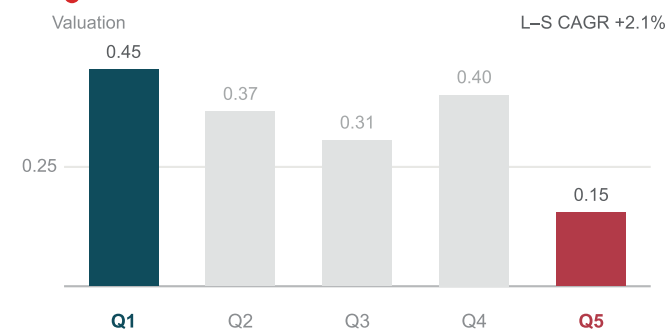
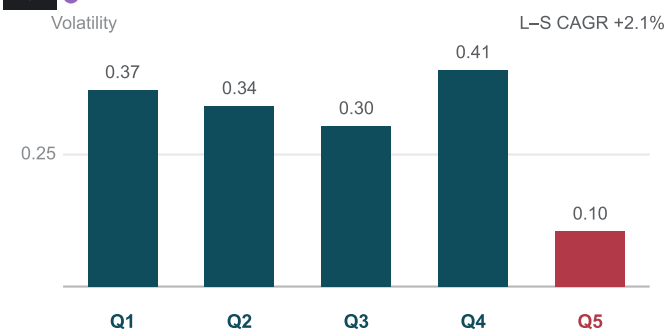
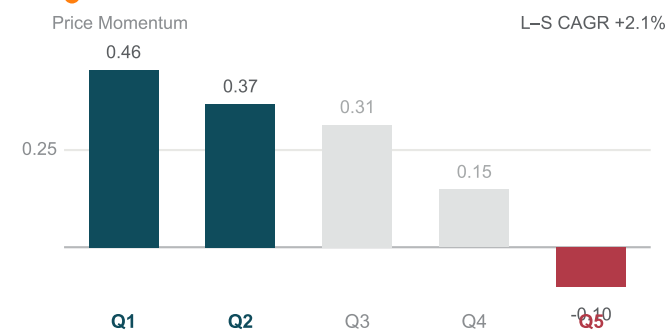


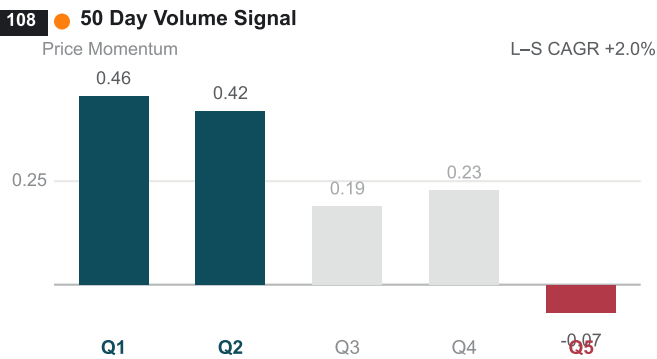
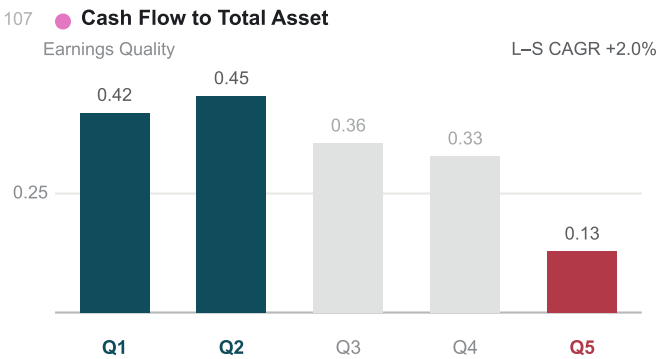
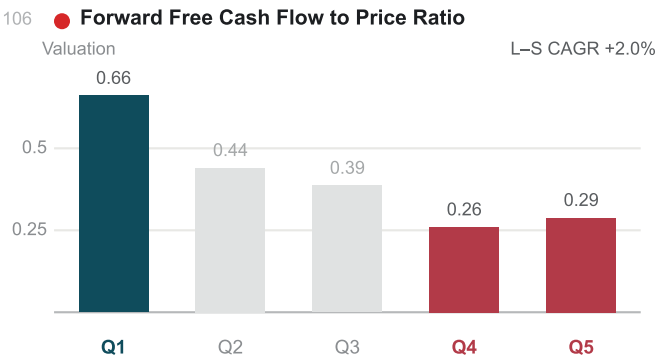
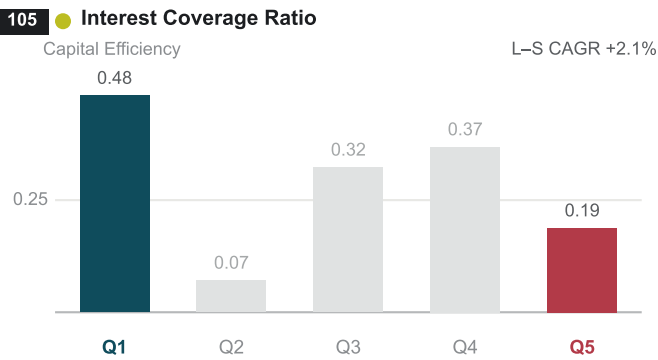
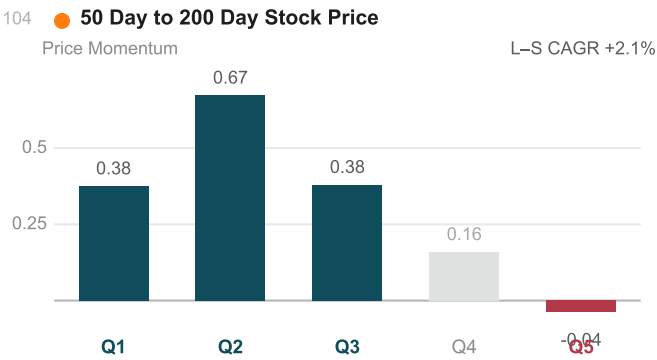




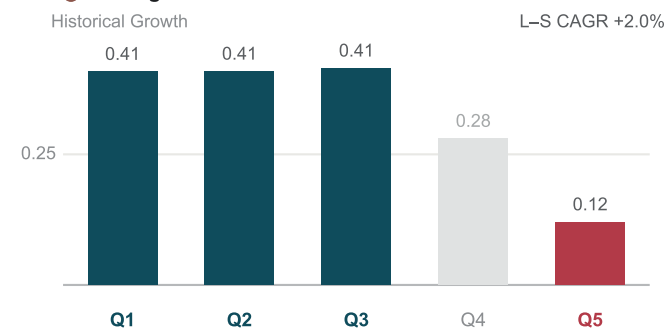
085 ● **Net Cash Flow to Enterprise Value**086 ● **Accrual Ratio - Balance Sheet**087 ● **Free Cash Flow to Enterprise Value**088 ● **Accrual Ratio - Cash Flows**089 ● **Max Daily Return in the Past 6 Months**090 ● **Working Capital to Assets Ratio**

091 ● **Adjusted Forward Earnings to Price**092 ● **Log of Unadjusted Stock Price**093 ● **Book to Price**094 ● **Percent Above 260-Day Low**095 ● **Pretax Income to Net Operating Assets**096 ● **Depreciation to Capex Ratio**

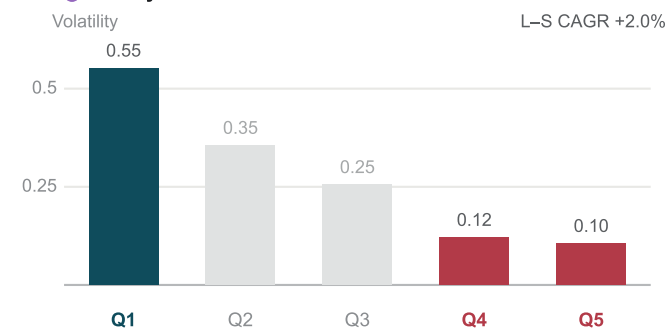
097 ● **15/36 Week Price Ratio**098 ● **Adjusted 6 Month Average Share Turn Over**099 ● **Solvency Ratio**100 ● **Cash to Total Assets**101 ● **60M CAPM Beta**102 ● **12M - 1M Price Momentum**



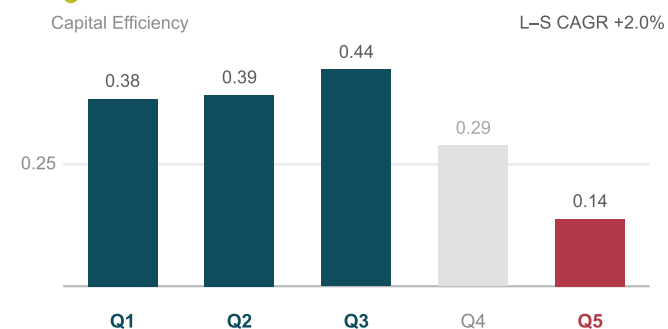
109 ● 3Y Chg in Sales Turnover



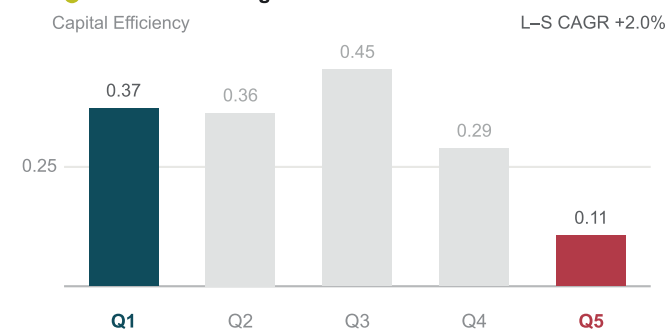
110 ● 90 Day Coefficient of Variation



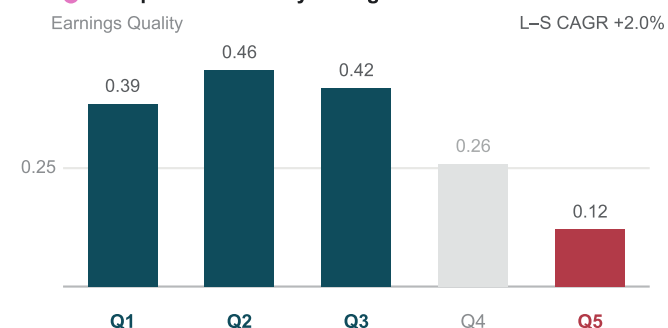
111 ● Debt to Assets Ratio



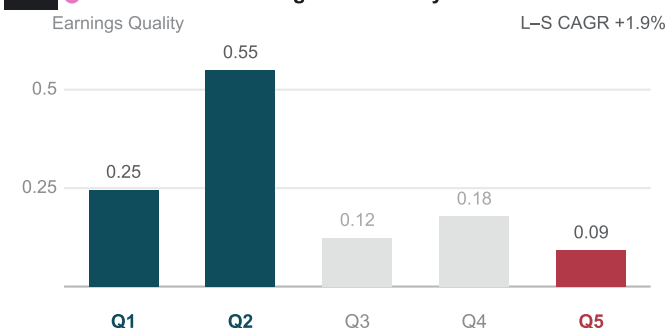
112 ● Pretax Profit Margin



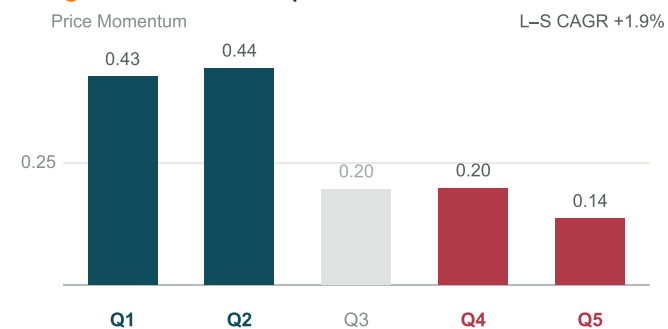
113 ● Unexpected Inventory Change



114 ● Year over Year Change of Inventory to Assets



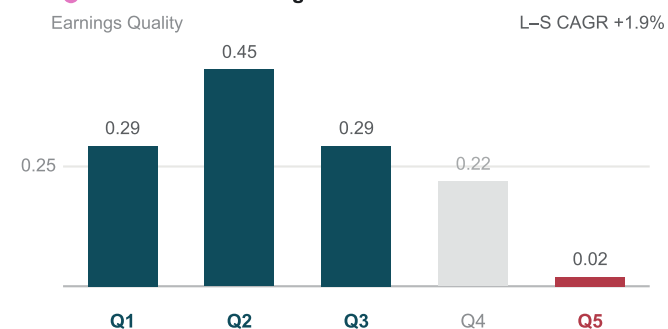
115 ● Institution Ownership Level



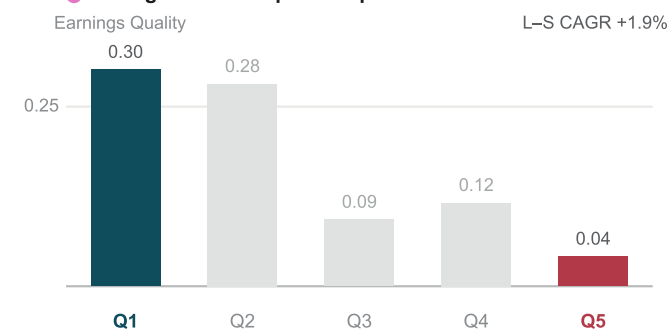
116 ● Dividend Growth



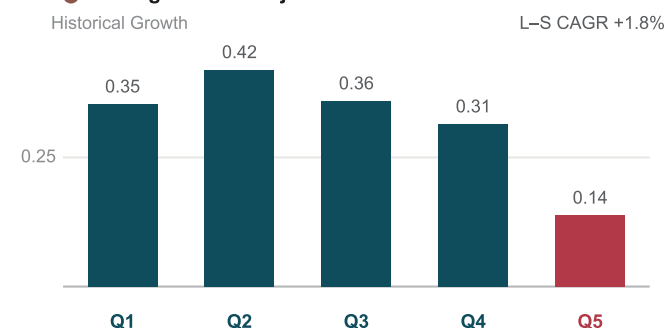
117 ● Year over Year Change of EPS to Sales



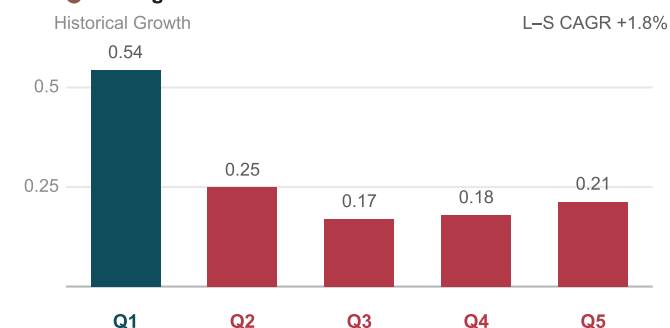
118 ● Change in TTM Depr. to Capex

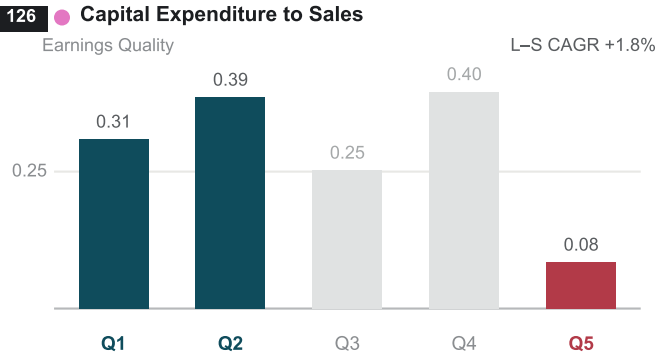
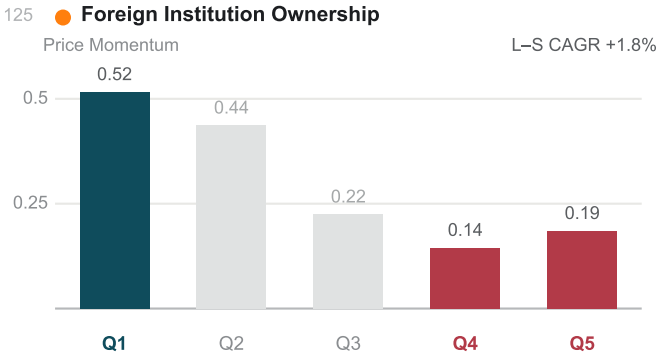
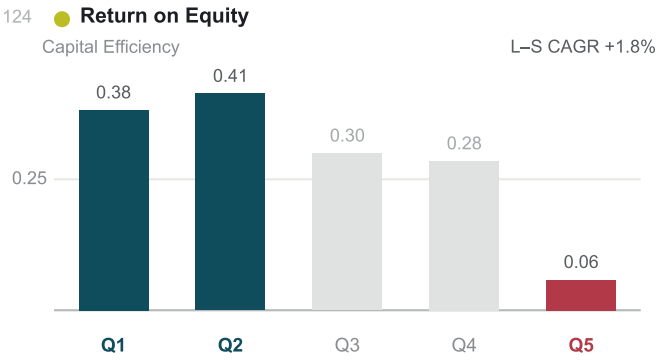
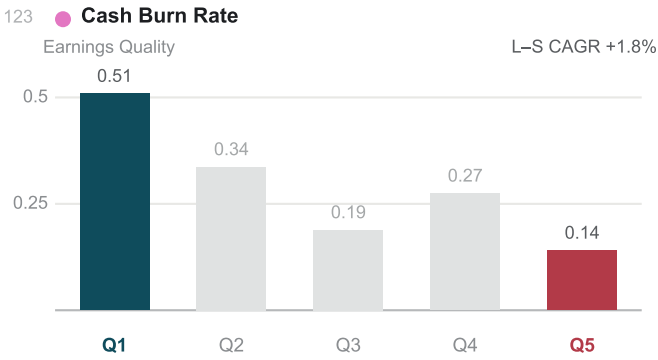
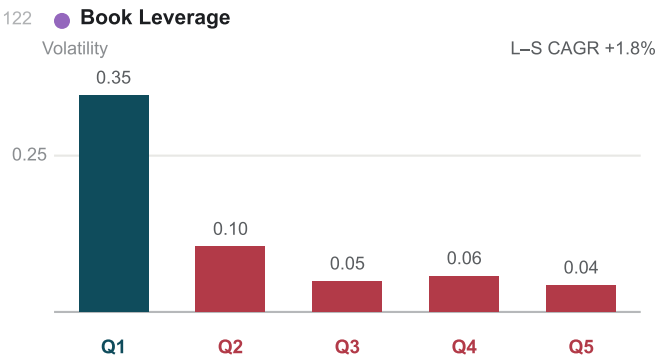
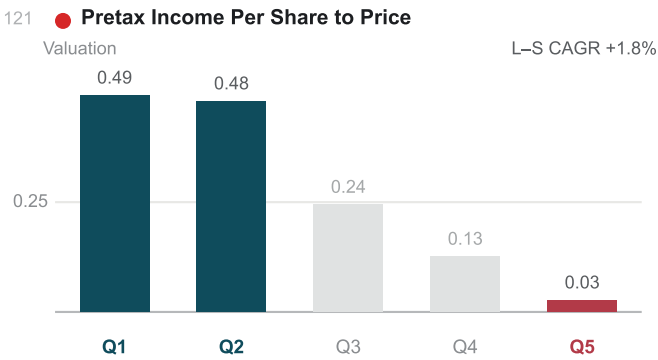


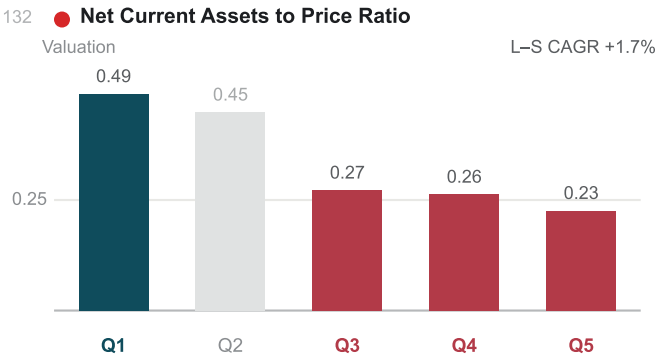
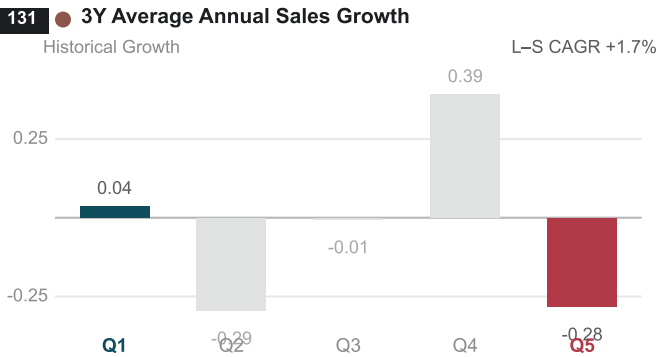
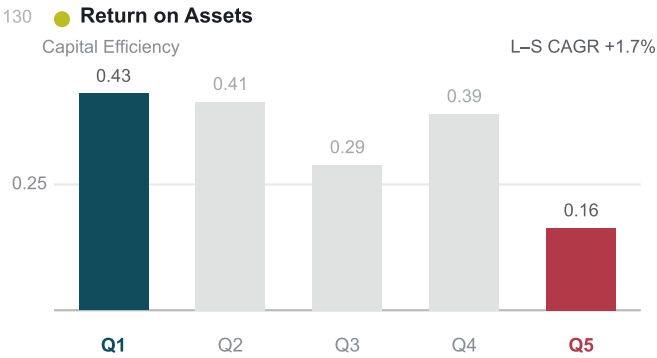
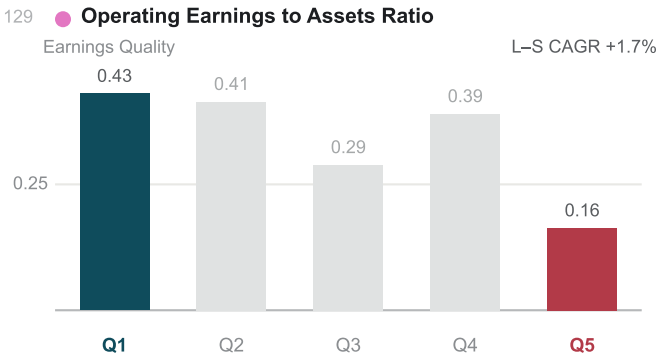
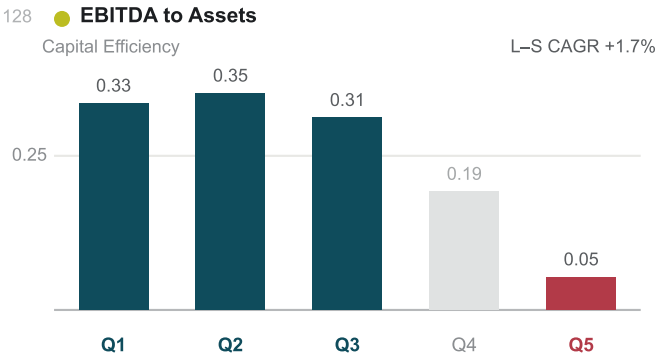
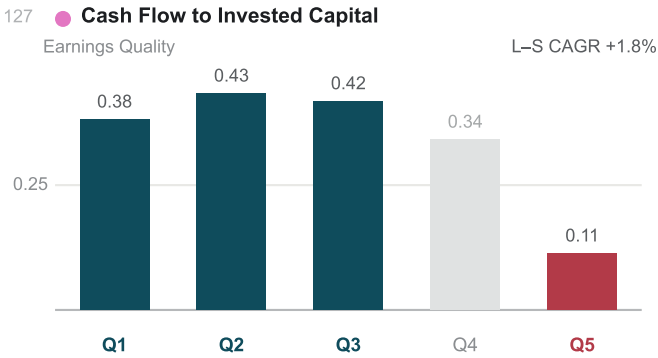
119 ● 3Y Chg in Asset Adjusted EPS

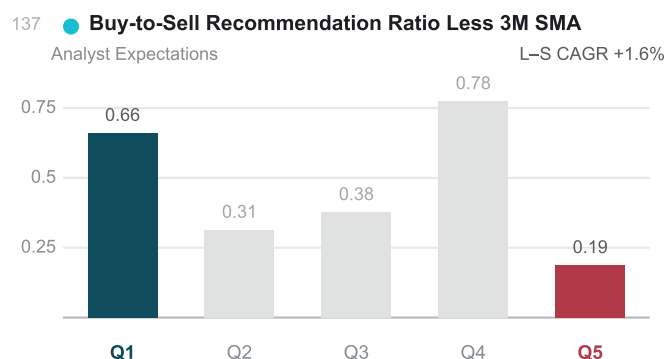
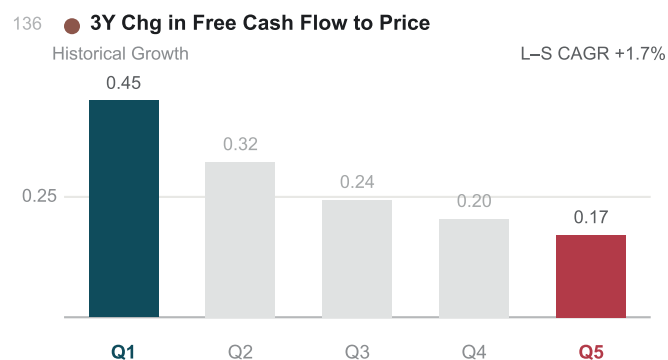
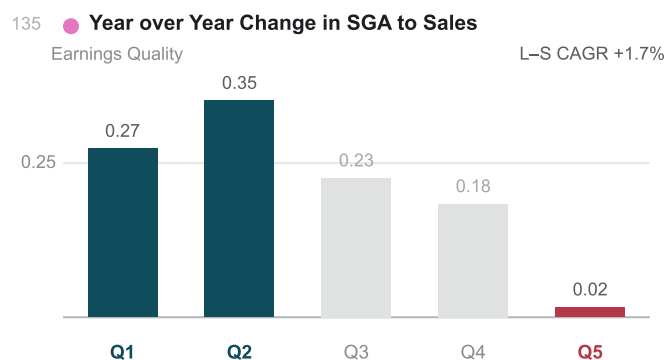
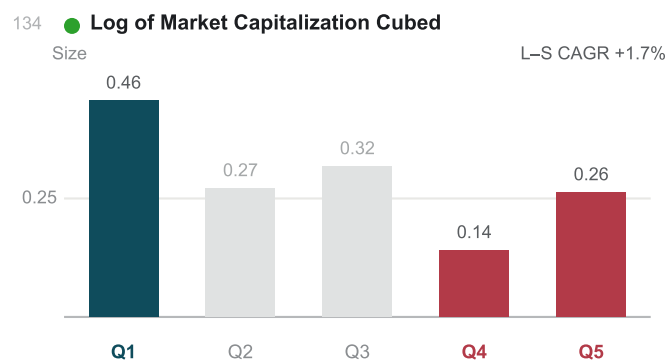
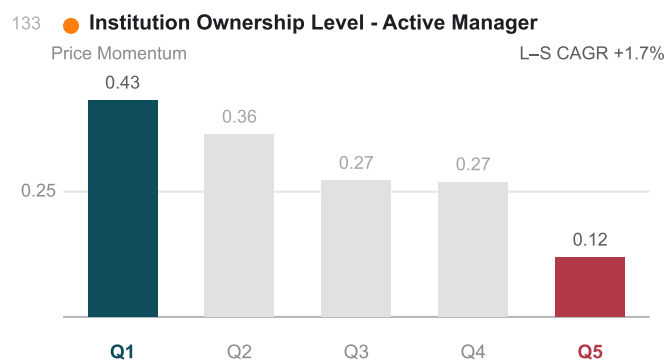


120 ● 3Y Chg in Cash Flow to Price





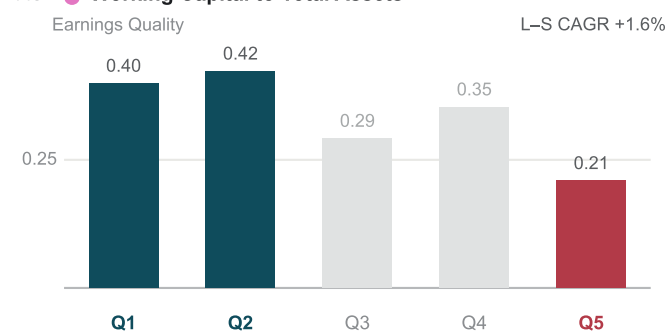




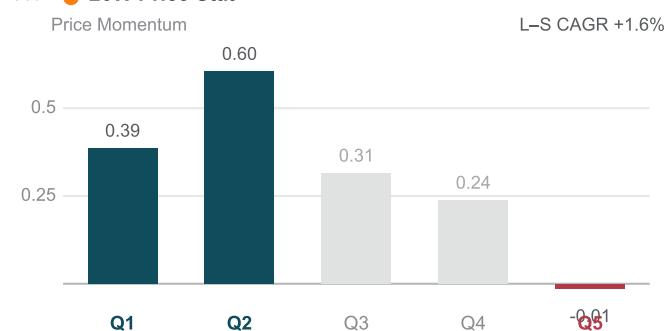
139 ● 3Y Chg in Asset Adjusted Free Cash Flow



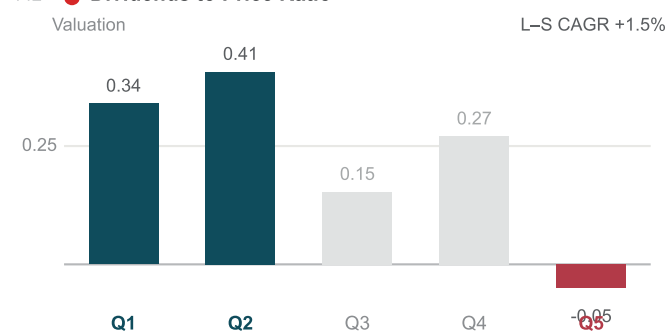
140 ● Working Capital to Total Assets



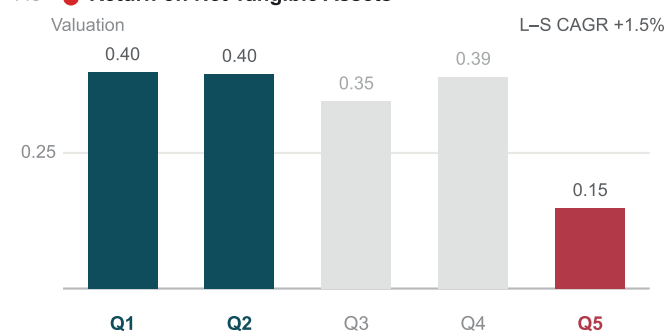
141 ● 26W Price Stat



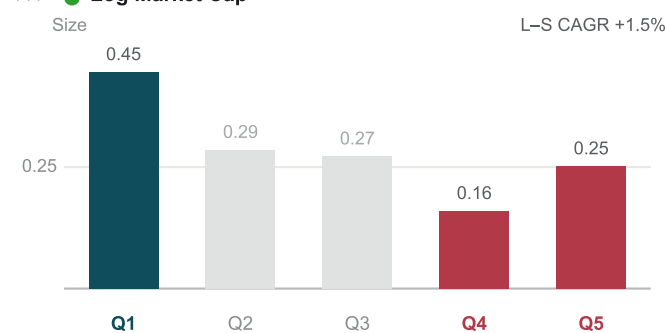
142 ● Dividends to Price Ratio



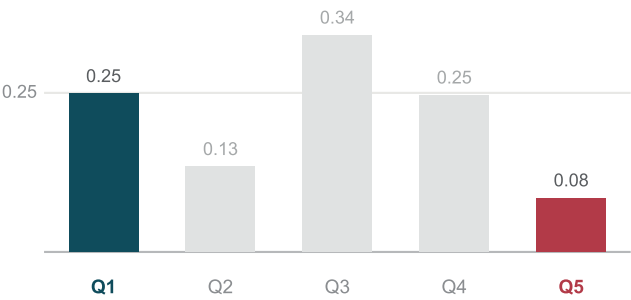
143 ● Return on Net Tangible Assets



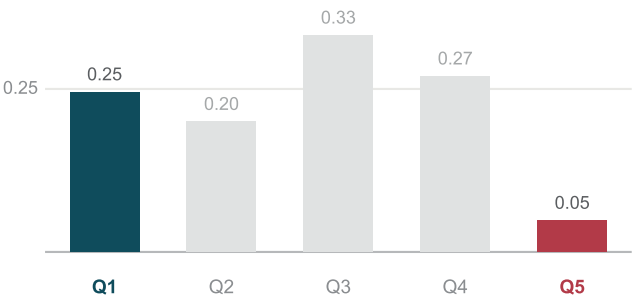
144 ● Log Market Cap



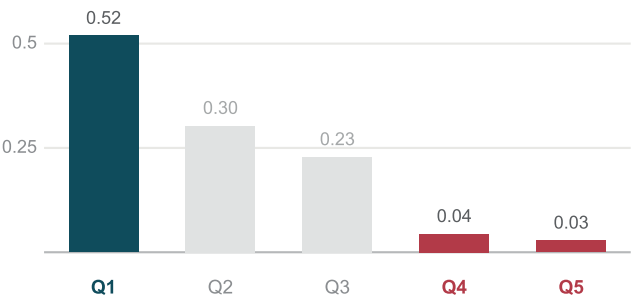
145 ● **Trailing 12-Month Receivables and Inventories/Trailing 12-M...**
Historical Growth L-S CAGR +1.5%



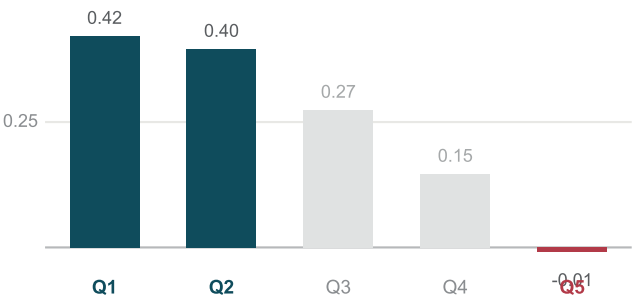
146 ● **Working Capital Turnover Ratio**
Capital Efficiency L-S CAGR +1.5%



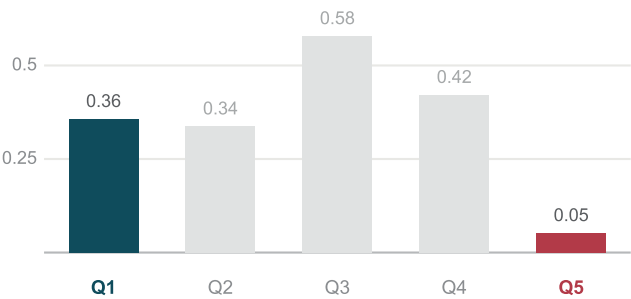
147 ● **EBITDA to Price**
Valuation L-S CAGR +1.4%



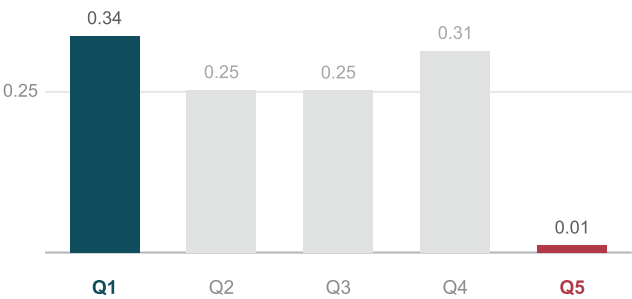
148 ● **Earnings to Price**
Valuation L-S CAGR +1.4%



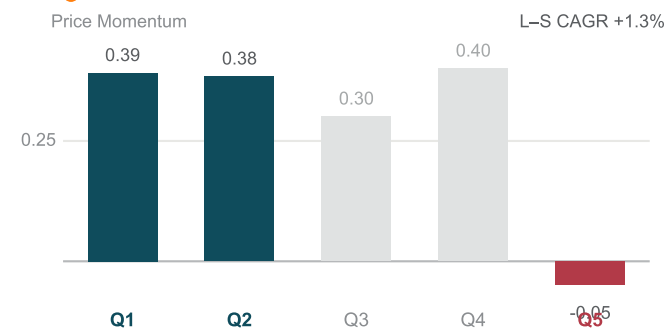
149 ● **3Y Chg in Cash Flow per Share**
Historical Growth L-S CAGR +1.4%



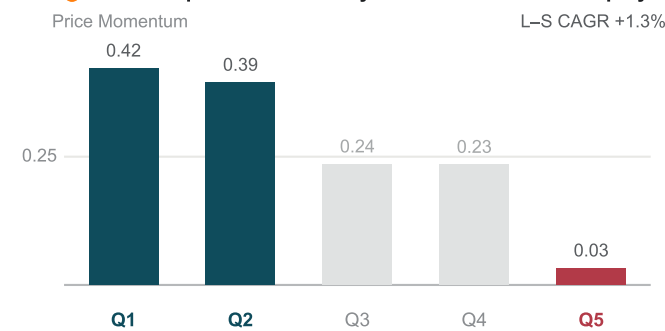
150 ● **Asset Growth - 1 Year**
Capital Efficiency L-S CAGR +1.4%



151 ● 6 Months Price Reversal



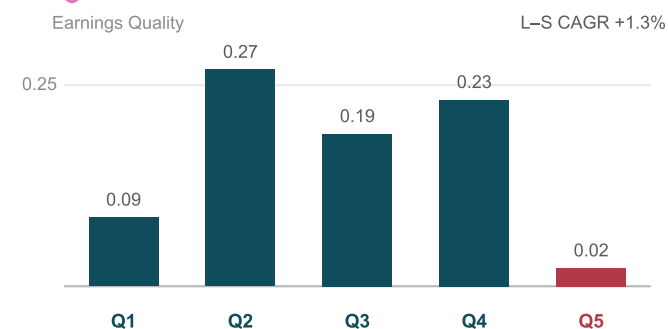
152 ● Ownership Breadth Stability of Institution Ownership by Buyers



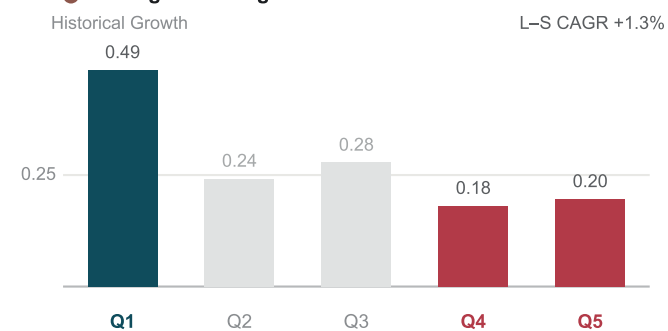
153 ● Forward Earnings to Price



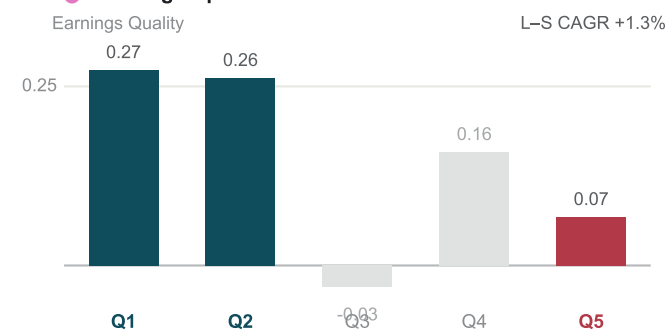
154 ● Cash Ratio



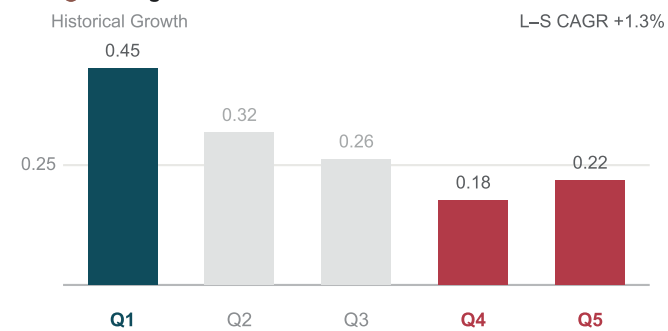
155 ● 3Y Chg in Earnings to Price



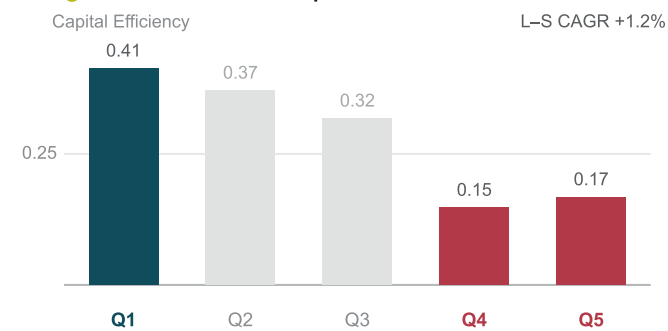
156 ● Working Capital to Sales



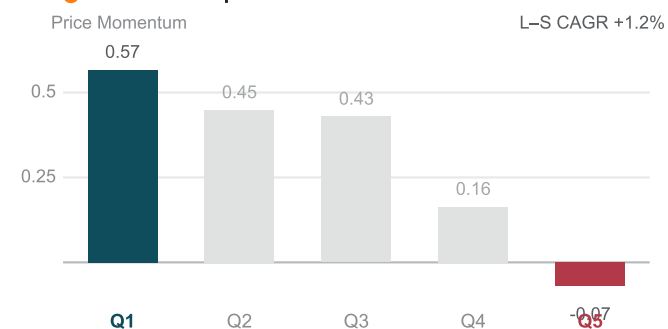
157 ● 3Y Chg in Sales to Price



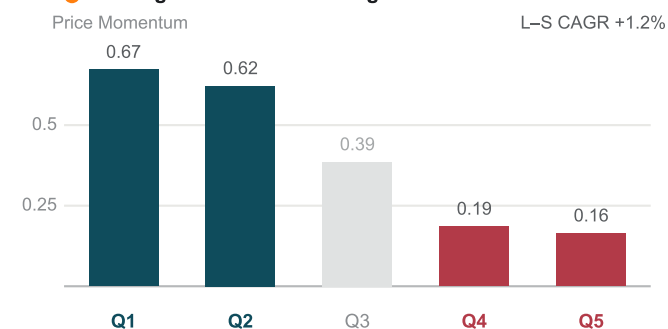
158 ● Return on Invested Capital



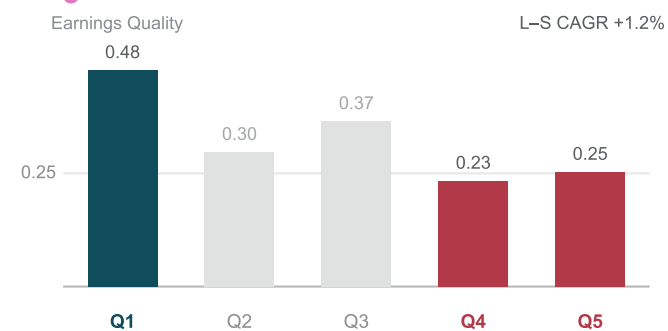
159 ● 60M CAPM Alpha



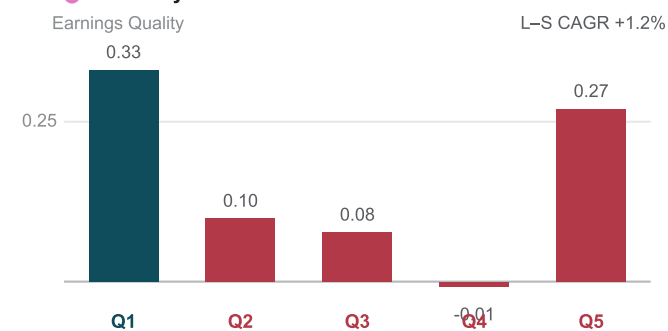
160 ● Closing Price to 52 Week High

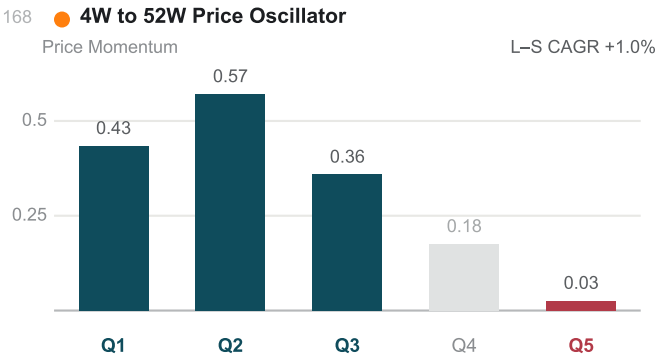
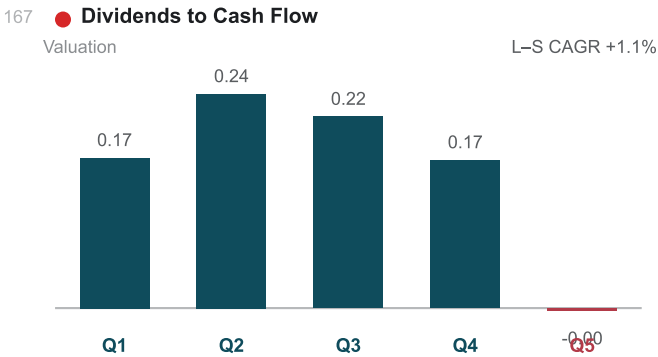
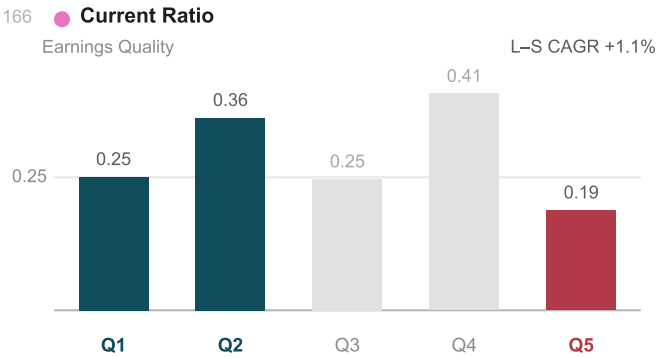
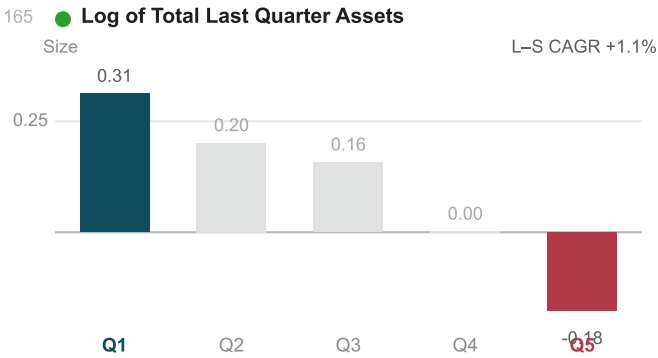
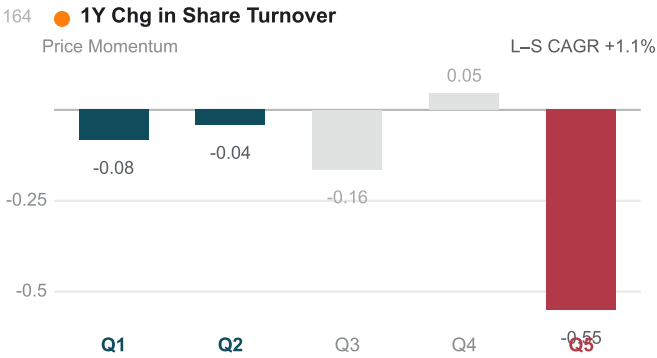
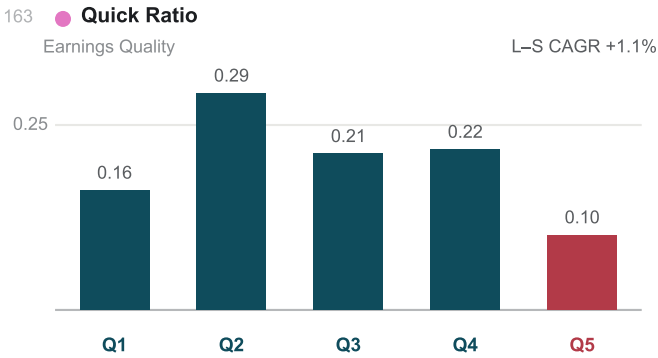


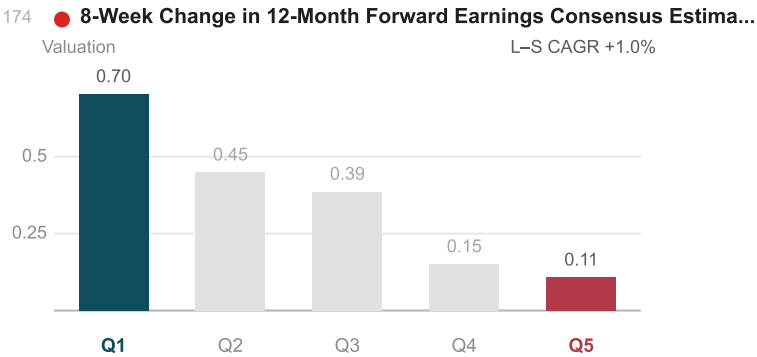
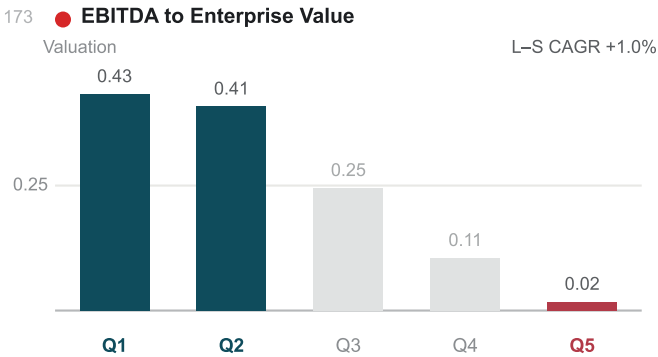
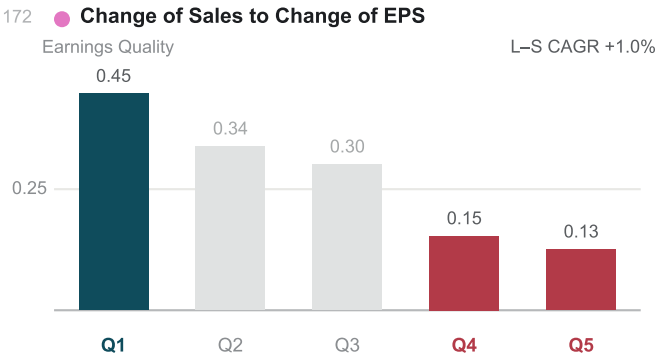
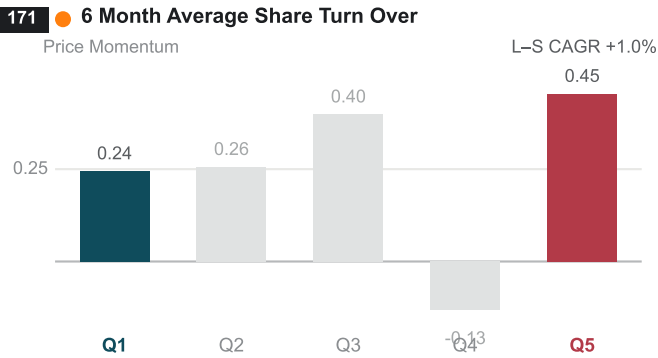
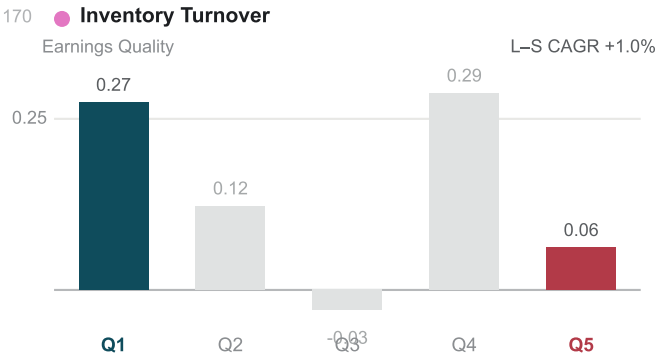
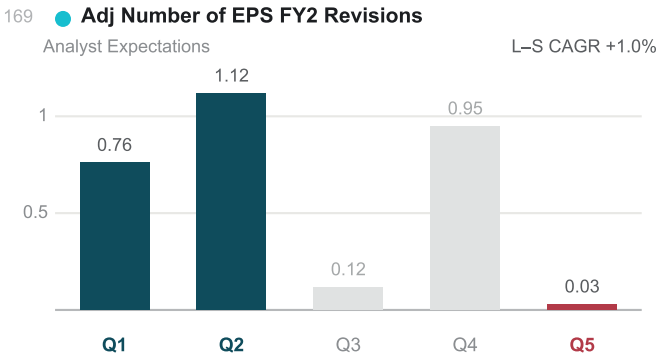
161 ● Cash to Sales



162 ● Inventory to Assets Ratio



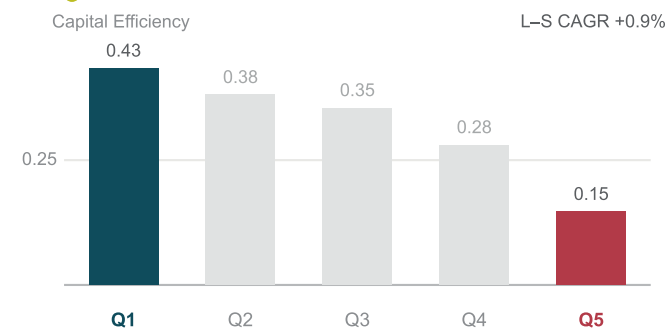




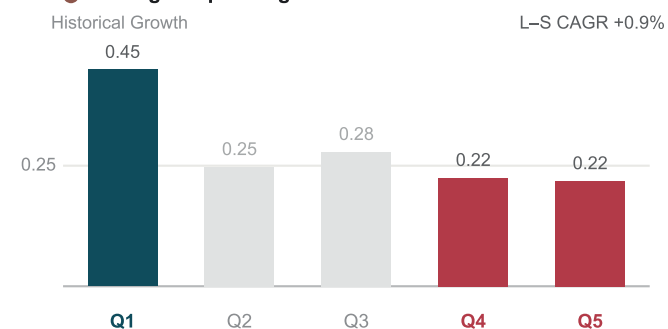
175 ● Degree of Financial Leverage



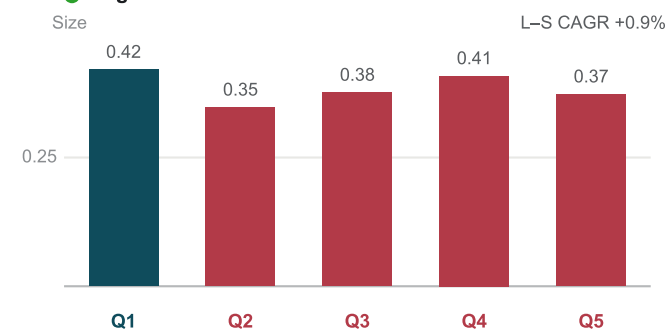
176 ● EBIT to Assets



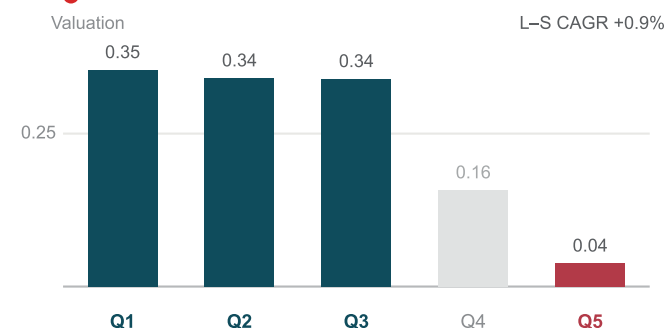
177 ● 1Y Chg in Operating Cash Flow to Price



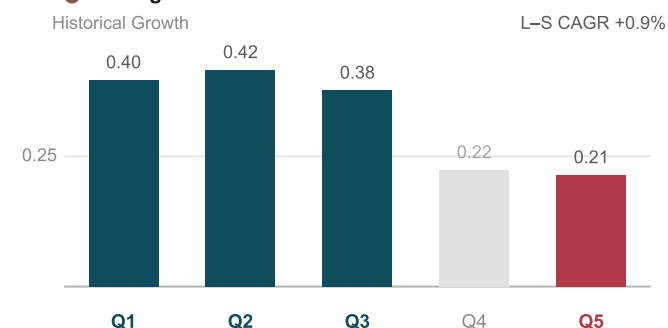
178 ● Log TTM Sales

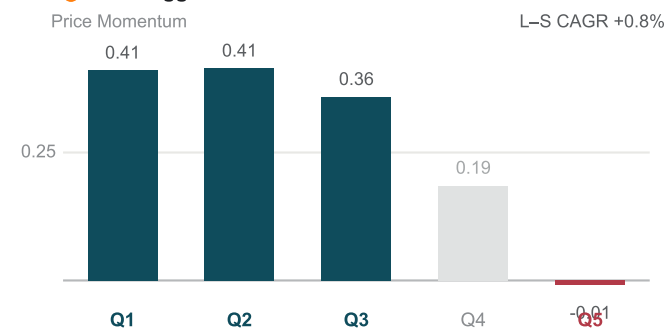
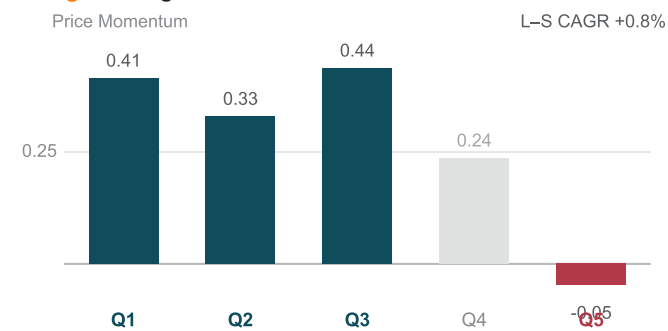
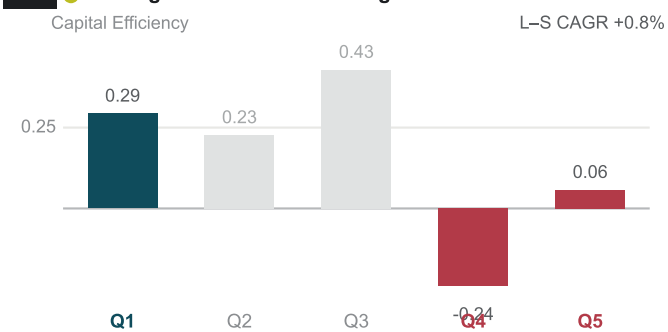
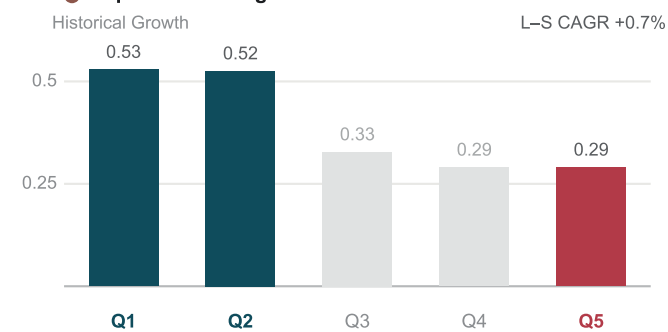
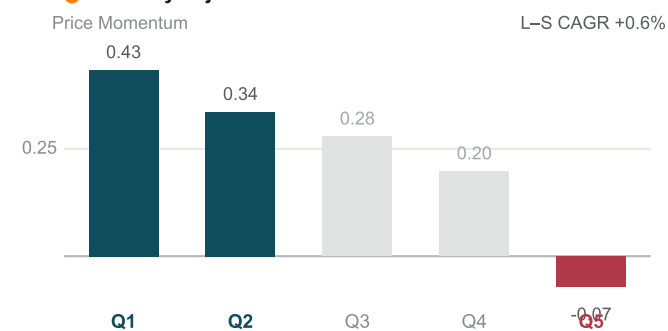


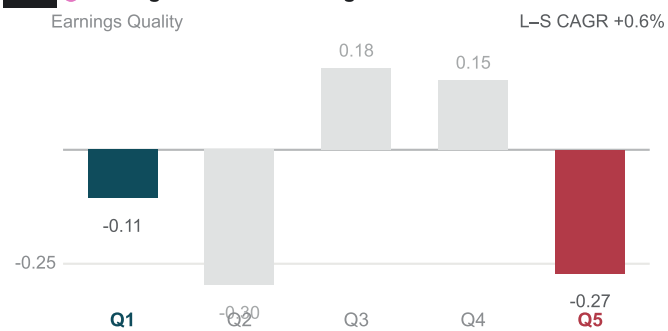
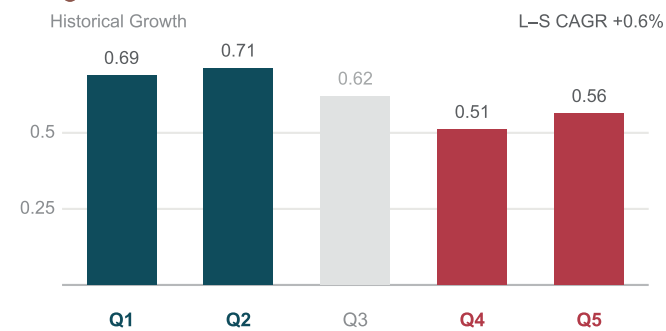
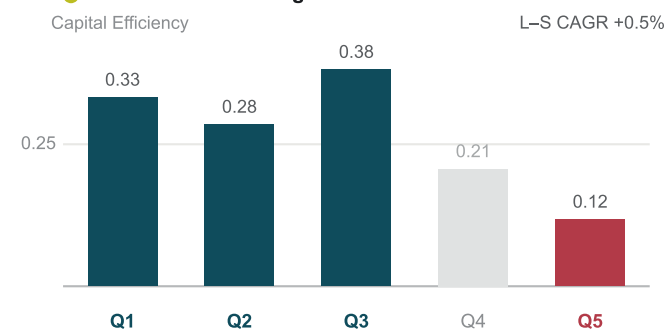
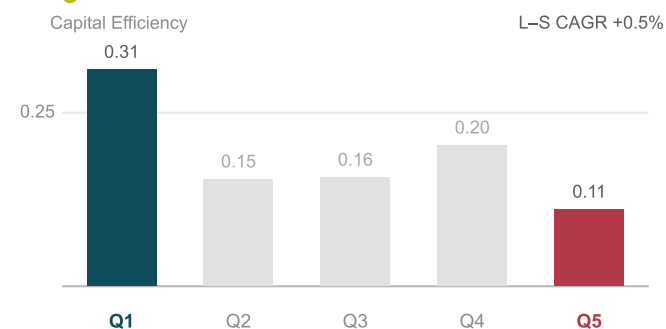
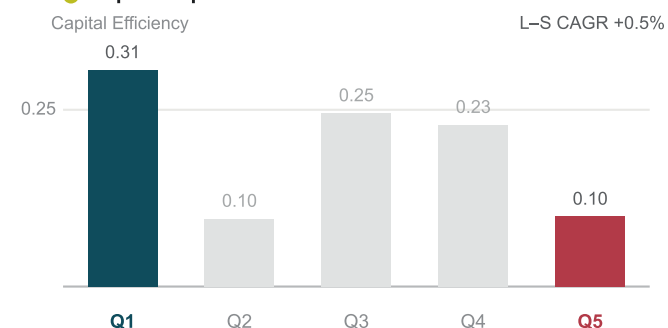
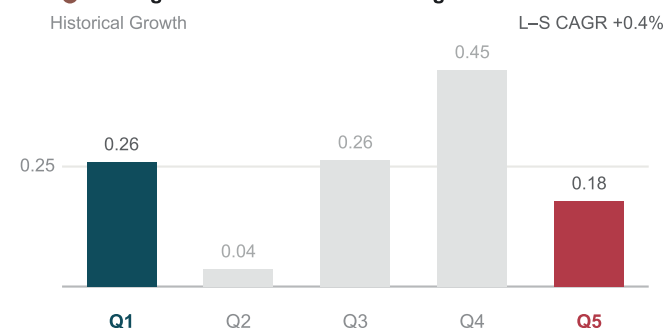
179 ● Normalized EP

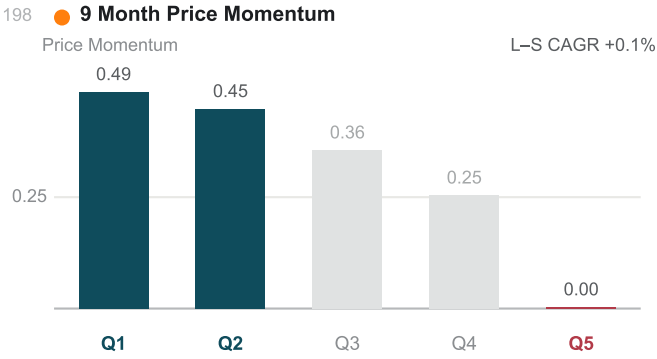
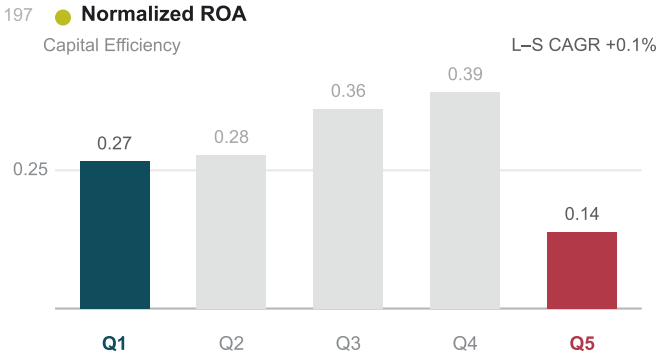
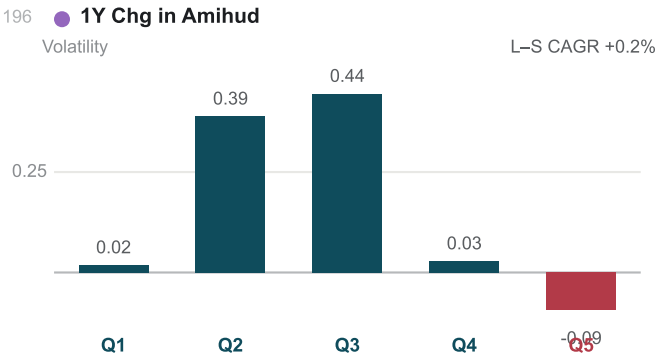
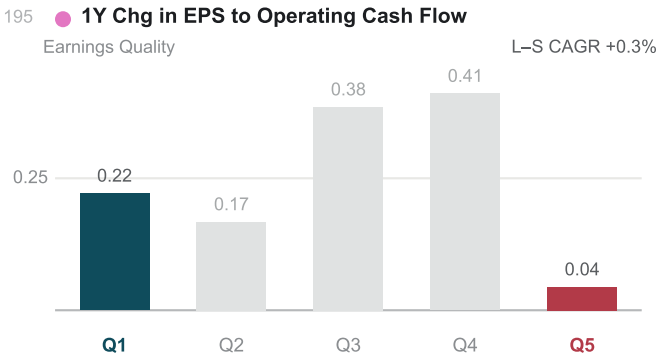
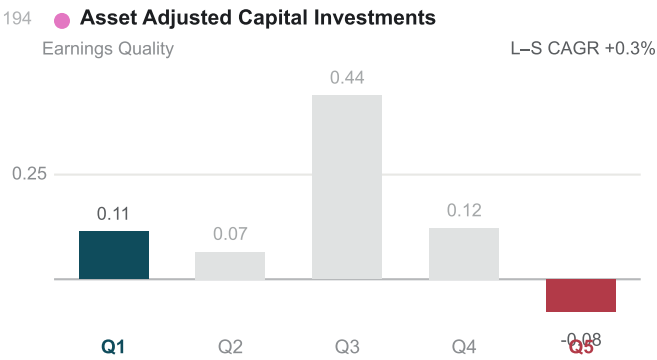
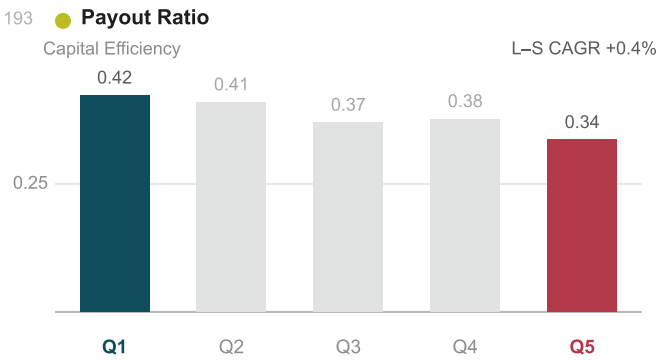


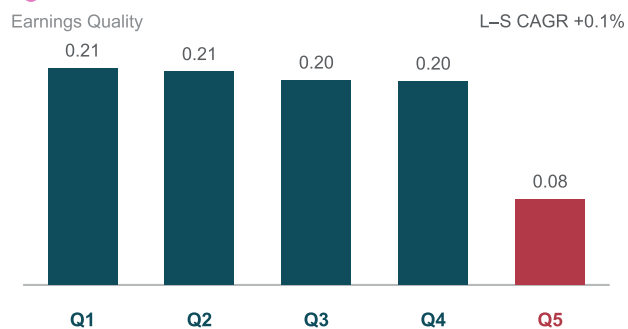
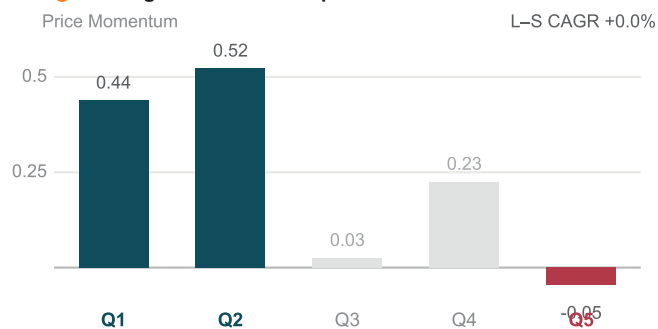
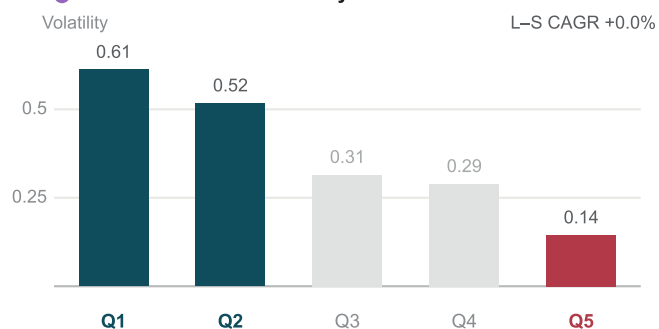
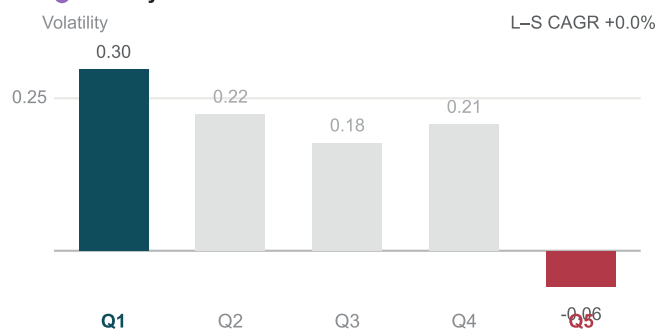
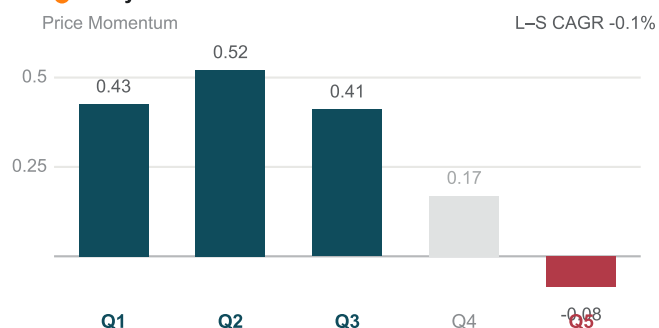
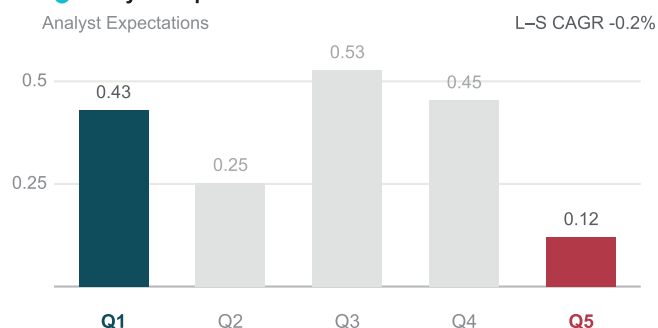
180 ● 1Y Chg in Free Cash Flow to Price



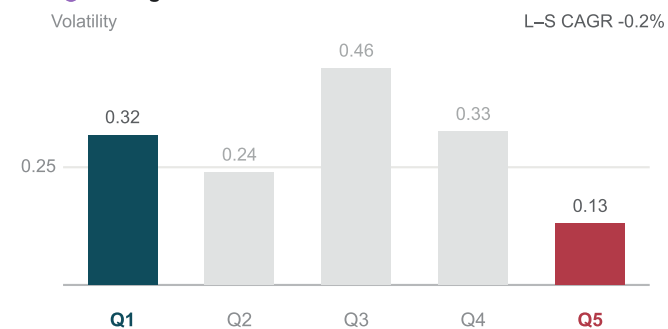
181 ● **39W Lagged Return**182 ● **52W High Low**183 ● **1Y Chg in Shares Outstanding**184 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**185 ● **Cash Conversion Cycle**186 ● **Volatility adjusted 12M return**

187 1Y Chg in Sales to Earnings**188 Sustainable Growth Rate****189 Year over Year Change of Total Debt****190 Normalized ROE****191 Capital Expenditure to Total Assets****192 Trailing 12-Month Overhead/Trailing 12-Month Sales**

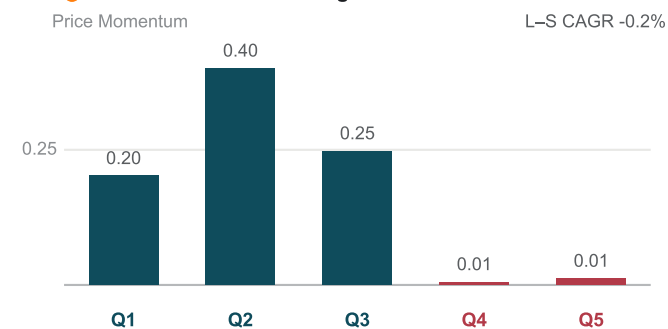


199 ● **Balance Sheet Accruals**200 ● **6M Chg in 12M CAPM Alpha**201 ● **1M Realized Price Volatility**202 ● **130 Day Minimum Return**203 ● **5 Day Price Reversal**204 ● **Analyst Dispersion for FY2 EPS**

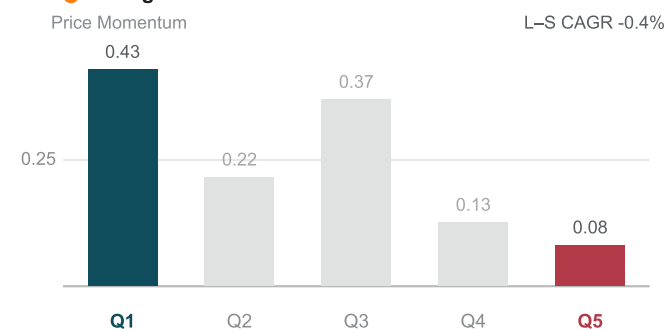
205 ● 1Y Chg in Debt to Assets



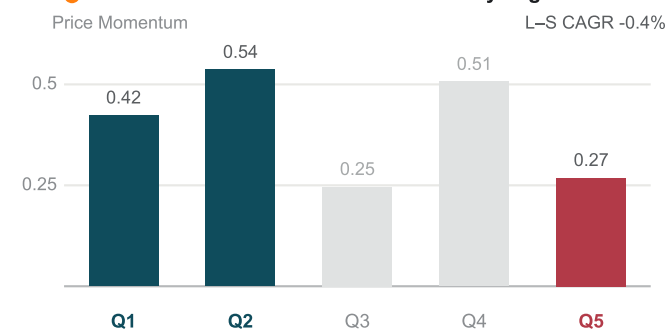
206 ● 26W Relative Price Strength



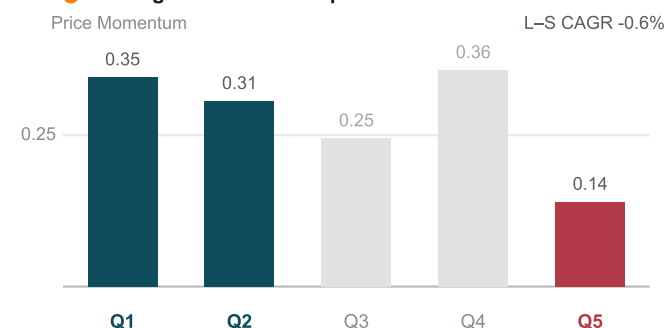
207 ● Change in Number of Holders



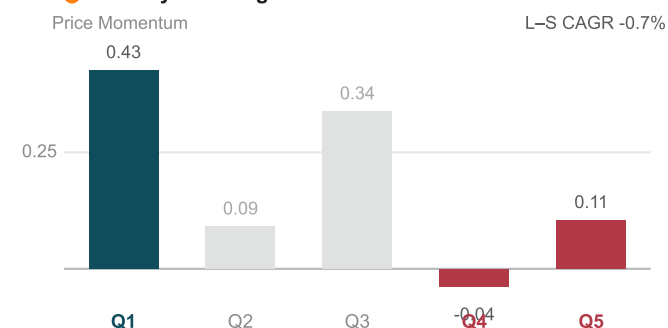
208 ● 52-Week Volume Price Trend with 20-Day Lag

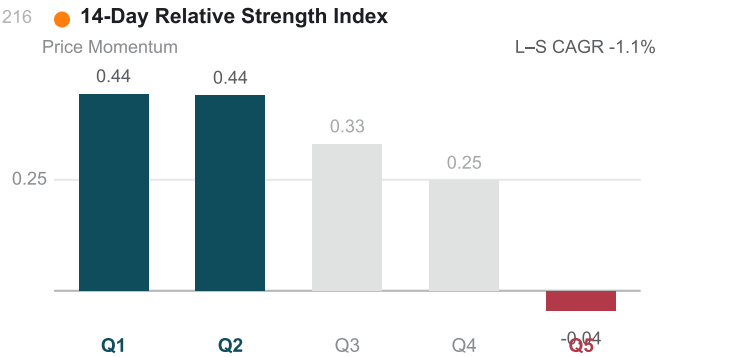
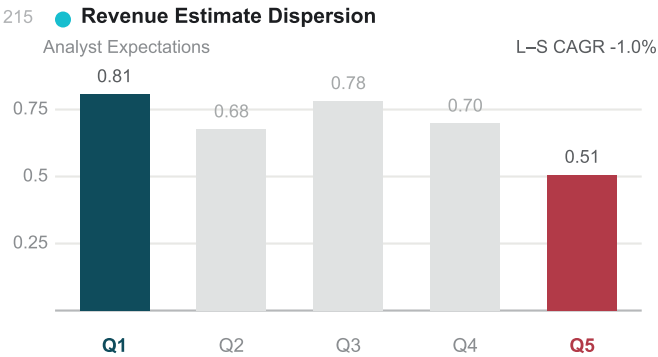
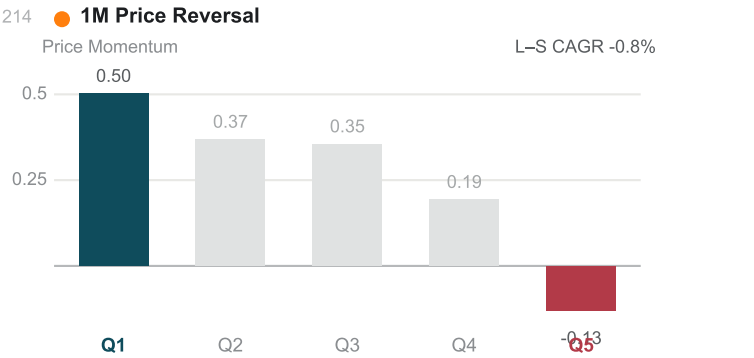
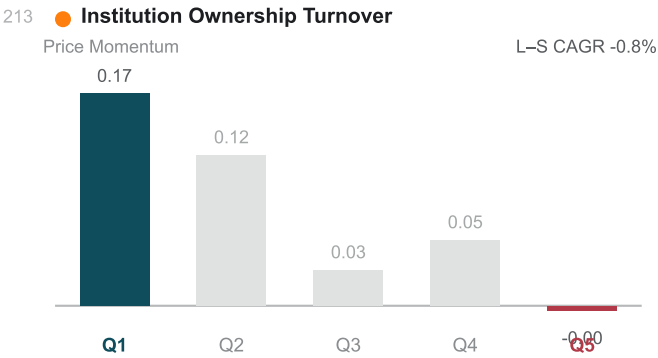
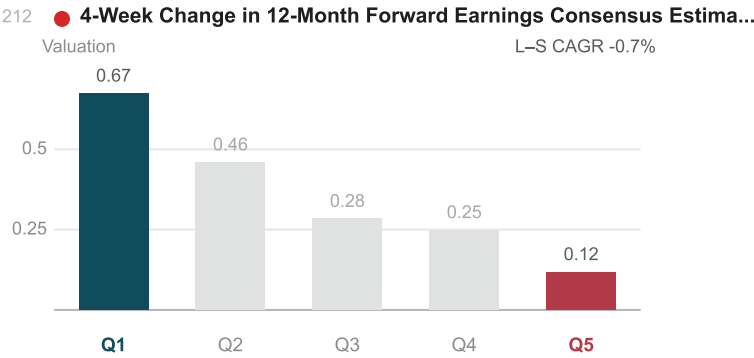
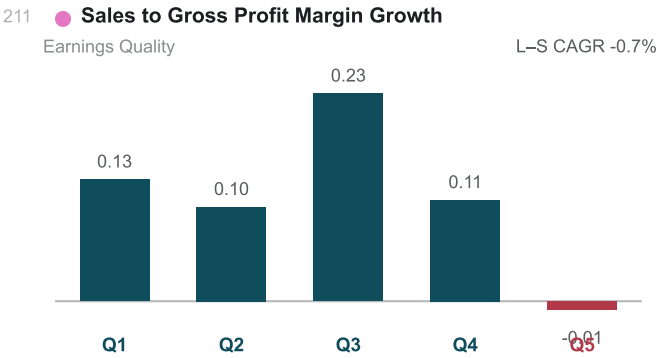


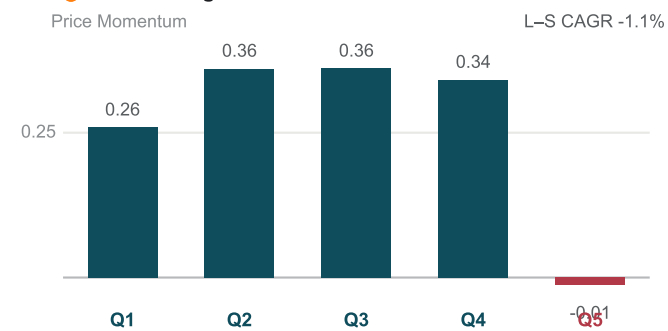
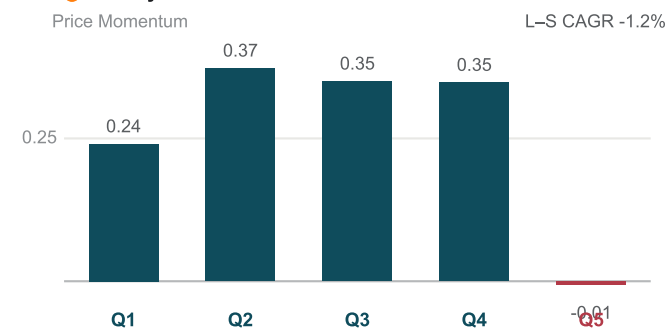
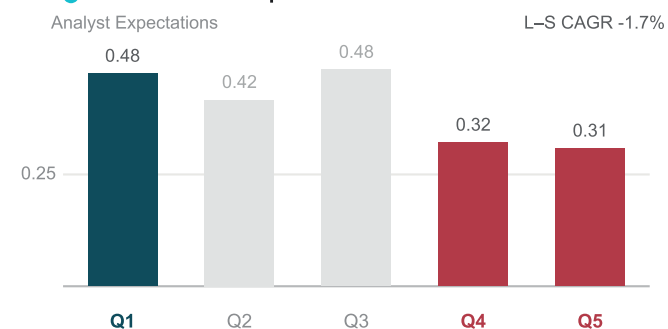
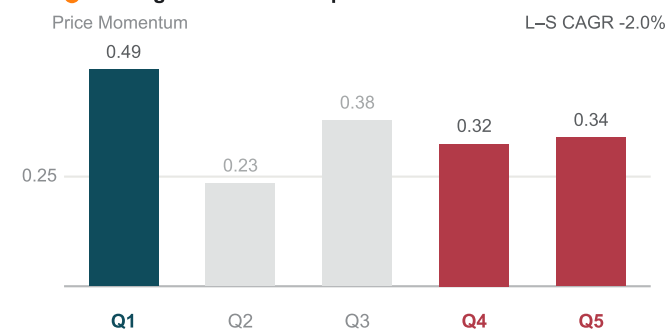
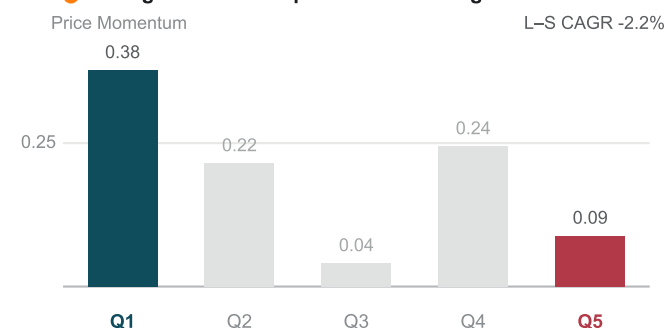
209 ● 6M Chg in 36M CAPM Alpha



210 ● Stability of Change in Number of Holders





217 ● **1M Price High - 1M Price Low**218 ● **20-Day Lane's Stochastic Indicator**219 ● **EBIT Estimate Dispersion**220 ● **6M Chg in 18M CAPM Alpha**221 ● **Change of Ownership Level - All Managers**222 ● **5-Day Money Flow/Volume**